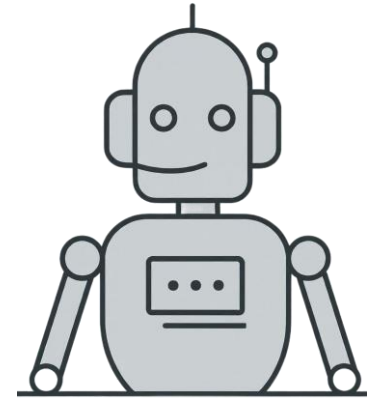


Building No Code Gen AI Workflows

Week 8: Project 1 - Introduction to Agents - MLS



Mike Lively

<https://olympus.mygreatlearning.com/courses/152926/modules>



Mike Lively

Founder QuantumAI
Trainer, IT Evangelist,
Developer
@mikelively-quantumai



17 Azure Cloud Certs



About Me

Father of Nine, USAF
Working on a PhD in GenAI
Teaching AI for Johns Hopkins & GK
Avid Hacker & Prompt Engineer
40 years programming, Keynote
Flautist
YMCA Gym Rat

PhD "ABD"





AB-100 Certification

By Michael L Lively 人

Certification Questions for AB-100 certifications.

Agentic AI Business Solutions Architect

Start a 10-question
adaptive AB-100
quiz with feedbac...

Give me a 20-
question AB-100
practice exam...

Quiz me on an AB-
100 topic of my
choice and identif...

Start an AB-
100 case study
and present the...

<https://chatgpt.com/g/g-6a584e1f27c48191bff33edef71bb326-ab-100-certification>



THANK YOU

FOR YOUR KINDNESS & SUPPORT
WITH LOVE FROM BERMUDA

Labs and Presentation Day 1 Download
https://github.com/eaglelandsonce/PL7008_Day1

Labs and Presentation Download
https://github.com/eaglelandsonce/PL7008_Day2



ICE Breaker

AI, using what you know about me, create a single-page cartoon résumé for an upcoming job interview. Depict my career journey, major achievements, and challenges overcome as a comic strip, featuring dynamic anime-style versions of me and an AI partner working together to conquer a seemingly impossible career challenge.

Share your Archtype and comic with the group. Were there any surprises.

MIKE LIVELY: CAREER QUEST

TURNING COMPLEXITY INTO MOMENTUM WITH AI, CLOUD, DATA, AND LEADERSHIP.

BUILDER. LEADER. TEACHER. PROBLEM SOLVER.

1 STRONG FOUNDATION

Mathematics, chemistry, engineering, physics—and lifelong curiosity.

2 LEADERSHIP IN SERVICE

Leadership forged in service. U.S. AIR FORCE CAPTAIN.

3 BUILDER & PROBLEM SOLVER

Build solutions in Java, C#, Python, and SQL.

4 QA ARCHITECT

Expanded enterprise testing from 1 to 40 teams.

5 DEVOPS TECH LEAD / SCRUM MASTER

Lead DevOps and Agile delivery across 20 scrum teams.

6 MANAGER OF TRAINING

Create onboarding and training for 500 employees.

7 GENAI CONSULTANT & EDUCATOR

WE DESIGN & DELIVER:

- Custom AI Agents
- Knowledge Graphs
- Pattern Recognition
- MMI & Vector Search
- Governance & Safety
- Measurable Impact

Microsoft Certified Trainer: AI systems and internal GenAI consultant.

TEACHING COMPLEX AI AI DESIGNING AI AGENTS SPEAKING & SHARING

Microsoft Copilot, GitHub Copilot, Google Gemini, OpenAI GPT-4o

DEVELOPING AI AGENTS: The future is Multia + AI Together.

8 THE QUEST: IMPOSSIBLE? NOT ANYMORE.

TOGETHER, WE TURN CHAOS INTO CLARITY.

IMPOSSIBLE CHALLENGE

AI + LEADERSHIP + CLOUD

Transforms complexity into business value.

MISSION ACCOMPLISHED!

ROLES	STRENGTHS	HIGHLIGHTS	MISSION	LET'S BUILD WHAT'S NEXT!
Developer - QA Architect - DevOps Tech Lead - Scrum Master Manager of Training - GenAI Consultant	- AI strategy - Cloud architecture - Automation - Data - Training - Agile leadership	23 certifications - Microsoft Certified Trainer - Founder, Quantum AI LLC - Community speaker and meetup leader	Help organizations adopt AI practically, responsibly, and at scale.	QR CODE Connect with me!

CURIOSITY STARTS THE JOURNEY. LEADERSHIP DRIVES IT. AI ACCELERATES IT. PEOPLE BENEFIT. — MIKE LIVELY

Agenda

Session Structure

- Review objectives, expectations, and session flow

Conceptual Recap

- Clarify key foundational concepts & additional explanation

Business Case Study

- Examine challenges in manual expense reimbursement
- Define an automated, policy-compliant, and auditable solution
- Build a three-workflow agentic system to:
 - Structure expense bills
 - Validate expenses against policy
 - Calculate approved and rejected amounts
 - Communicate reimbursement decisions

Guided Hands-On Demonstration

- Process bill images with GPT-4o vision to extract, categorize, and total expenses.
- Build an audit and self-correction loop to identify and resolve errors.
- Expose reimbursement workflows as MCP tools and connect the AI agent through SSE.
- Retrieve policy data, calculate reimbursement decisions, and send the final email.

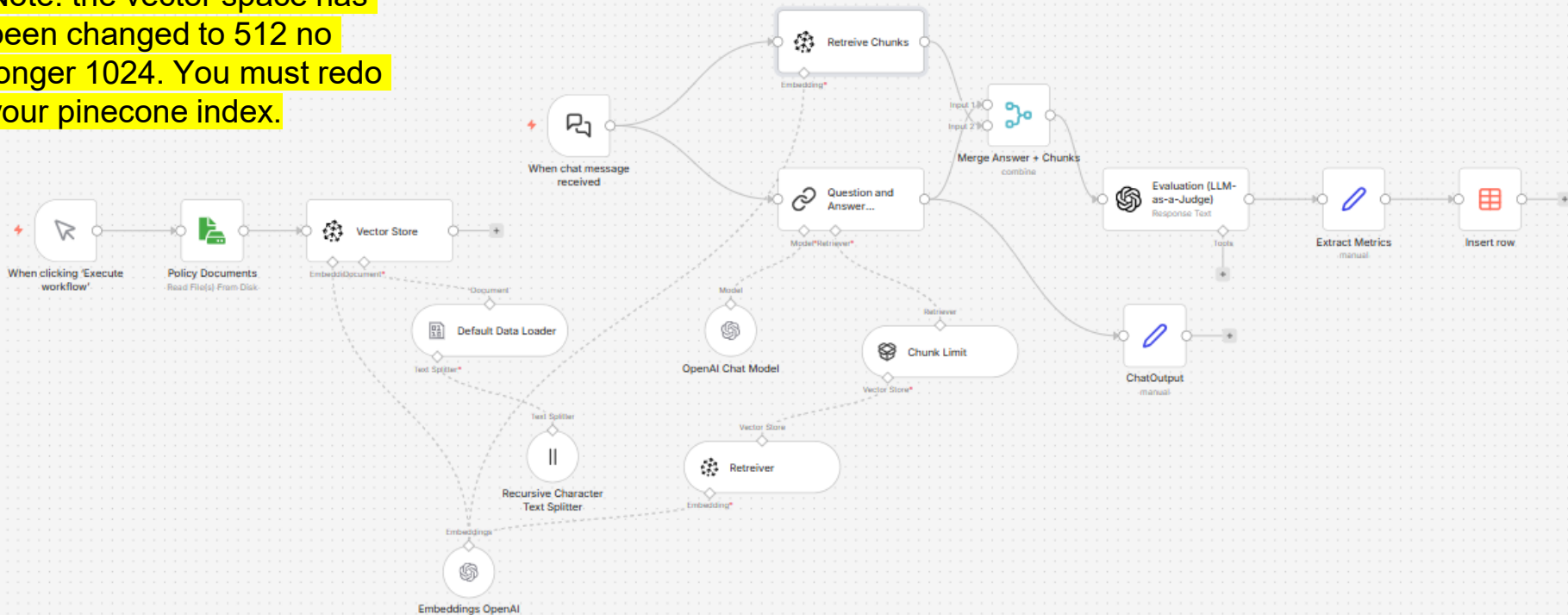
Questions and Answers

Address follow-up and parked questions



Fixed Week 5 Solution

Note: the vector space has been changed to 512 no longer 1024. You must redo your pinecone index.



https://github.com/eaglelandsonce/FixedWeek5Solution_n8n

Step 5.2 - Configuring the New Index in Pinecone

- ❖ Go to Pinecone dashboard → Navigate to **Database**, go inside 'indexes' → Click **Create Index**.
- ❖ Configure: **Example Index Name:** **insurancepolicy**, **Embedding Model:** **OpenAI text-embedding-3-small**, **Dimension:** **512** (Leave remaining settings as default)

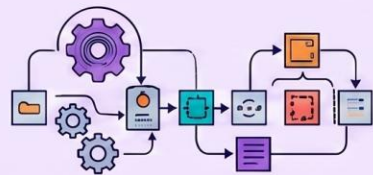
The screenshot shows the Pinecone dashboard interface. The top navigation bar includes 'Pinecone', 'Great Learning', 'Chatbot application', 'Database', 'Marketplace', 'New', 'Docs', 'Settings', 'Get help', and a user profile 'PK'. The left sidebar contains links for 'Get started', 'Database', 'Quickstart', 'Indexes (3)', 'Backups', 'Assistant', 'Inference', 'API keys', and 'Manage'. The main content area is titled 'Configuration' and shows three options for the embedding model: 'text-embedding-3-large' (Most powerful OpenAI embedding model), 'text-embedding-3-small' (Most cost effective OpenAI embedding model), and 'Manual configuration' (Customize settings for your embedding model). The 'text-embedding-3-small' option is highlighted with a red box. Below the model selection, a table shows the index configuration for 'text-embedding-3-small':

Index configured for text-embedding-3-small:	
Modality	Text
Vector type	Dense
Max input	8,191 tokens
Dimension	512
Metric	cosine

To the right of the table, a section titled 'Manage your own embeddings' provides instructions: 'This model does not support integrated inference. To manage your own embeddings, follow these steps: 1. Create an account: Sign up and obtain an OpenAI API key. 2. Embed: Generate vector embeddings for your data. 3. Upsert: Insert the generated embeddings into your index.' At the bottom left, a 'DATABASE USAGE' section shows 'RUs 7 / 1M', 'Storage 0.016GB / 2GB', and 'WUs 279K / 2M', with an 'Upgrade now' button. At the bottom right, a 'Create index' button is highlighted with a red box, next to a 'Cancel' button. A status bar at the bottom indicates 'You are currently on the Starter plan'.

AI Agents: Navigating the Shift from Automation to Autonomy

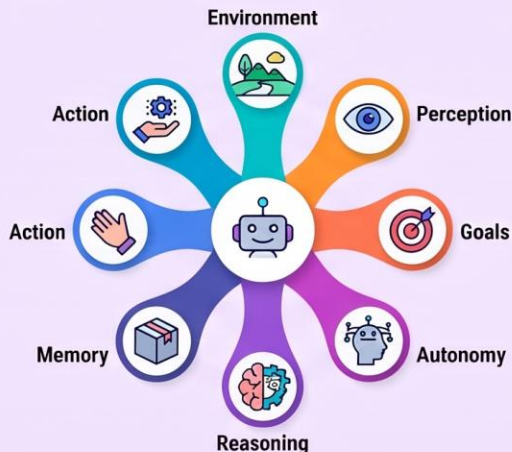
The Anatomy of Autonomy



TRADITIONAL AUTOMATION

Beyond Traditional Automation

Agents autonomously reason and plan rather than following rigid, predefined "if-then" recipes.



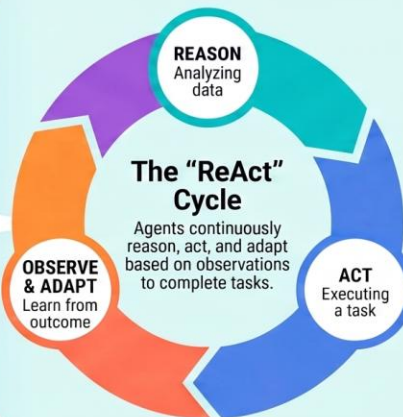
The GenAI Advantage

Modern agents use natural language proficiency and tool utilization to solve unforeseen problems.

Seven Core Pillars

Behavior is driven by environment, perception, goals, autonomy, reasoning, memory, and action.

Strategic Business Implementation



Analysts



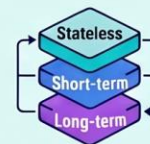
Customer Support assistants



Sales lead qualifiers

Applied Enterprise Personas

Purpose-built agents act as Analysts, Customer Support assistants, and Sales lead qualifiers.



Memory Management

Organizations must balance stateless, short-term, and long-term storage for personalization and privacy.

ACCELERATION OF AI AGENT CAPABILITIES



Siri (2011)

Mainstream voice-controlled interaction and virtual assistance.



GPT (2018-Present)

LLMs providing powerful reasoning and generative capabilities.



AutoGPT (2023)

Pioneering autonomous goal-setting and execution using LLMs.

Crossing the Embodiment Threshold

Digital coordination via APIs is only half the story. Embodiment introduces real sensor data, physics constraints, and irreversible consequences.

Digital Realm

- Discrete state spaces.
- Instant execution.
- Easily reversible.
- Cheap to reset/retry.
- Deterministic simulation.

Assistant

Physical Realm

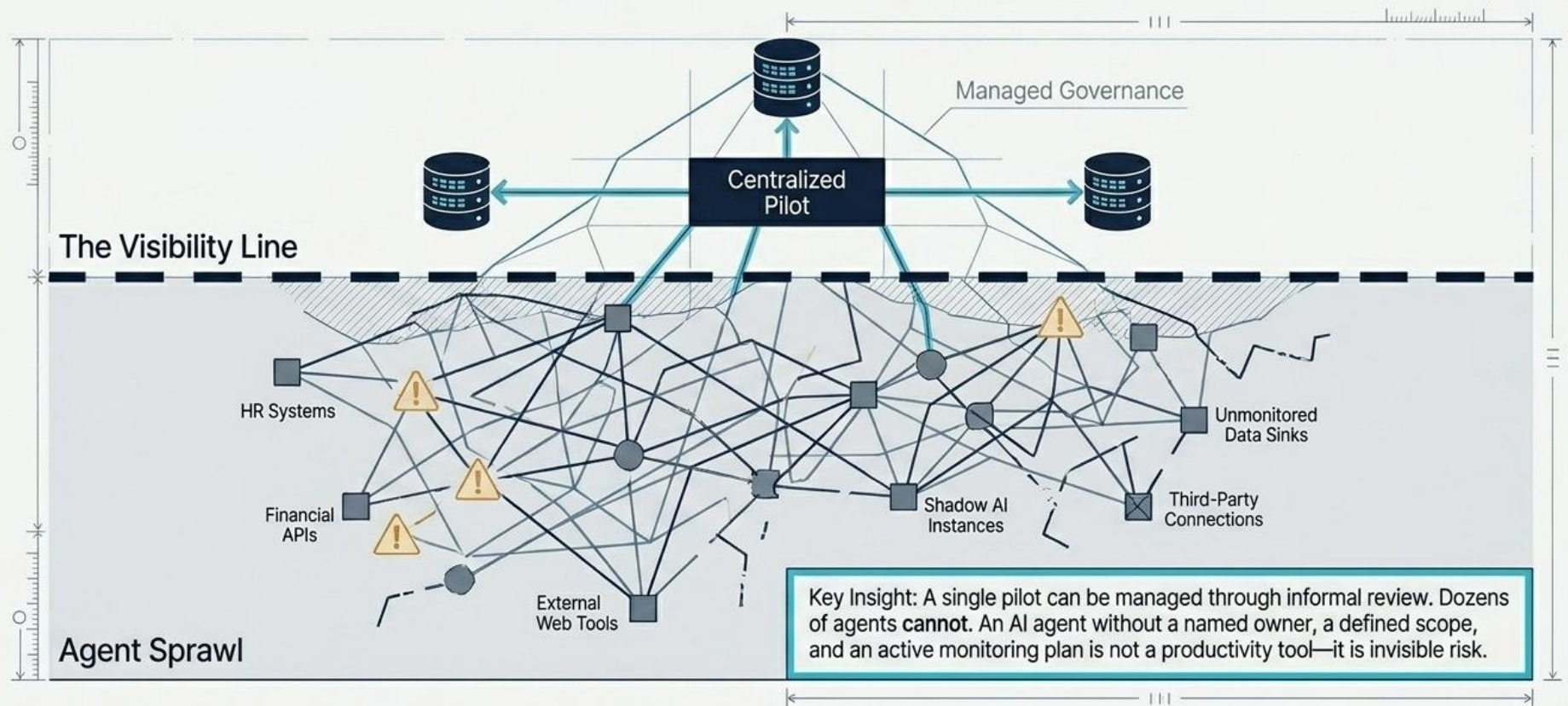
- Continuous dynamics.
- Actions unfold over time.
- Often irreversible.
- Costly errors (equipment damage, injury).
- Stochastic, noisy processes.

Agent





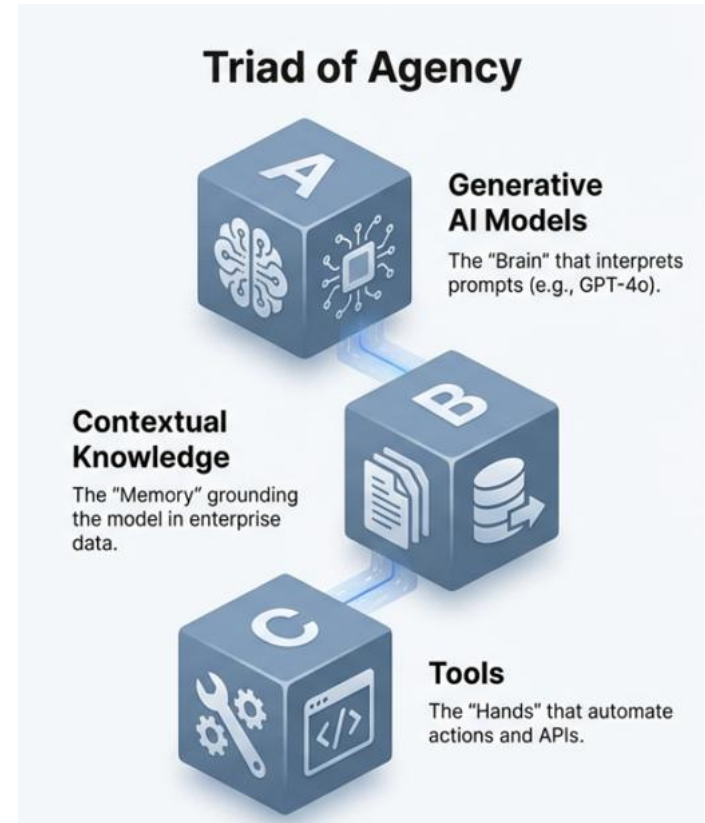
Without centralized visibility, agent sprawl becomes an unmanaged liability.



What's an Agent

Simple Definition

An AI agent is a system that can **perceive, reason, and act** toward a goal with autonomy.



https://olympus.mygreatlearning.com/gl_coding_labs/aB3r93yA

Market_Research_Agent

Demo

Personal / Market_Research_Agent + Add tag

0 / 1

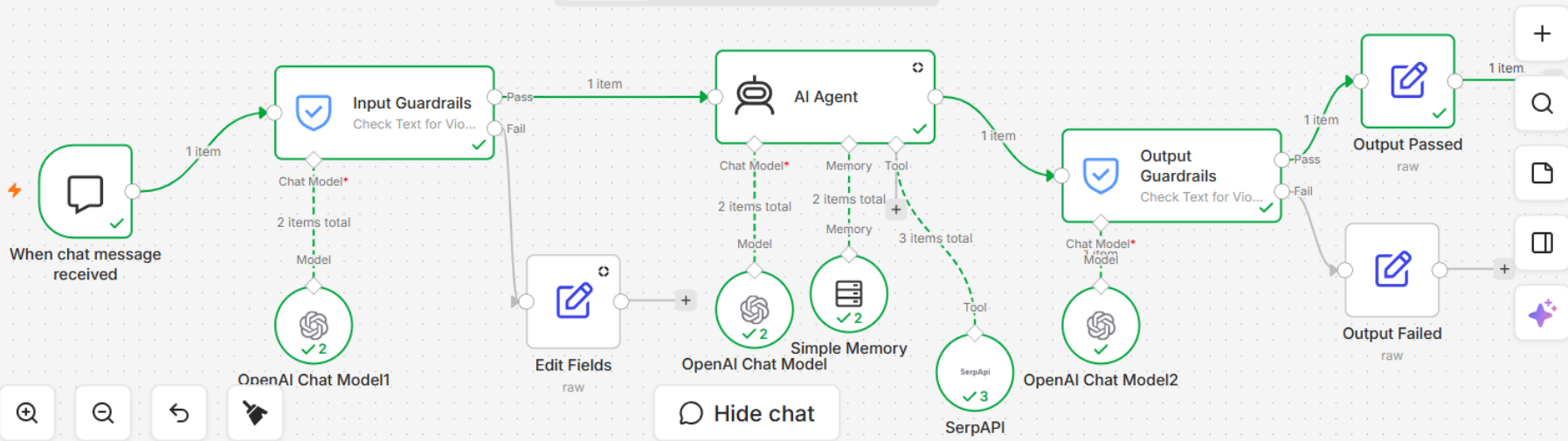
Publish



Editor

Executions

Evaluations



Chat

Session: aa584...

Logs

Clear execution



Success

Input

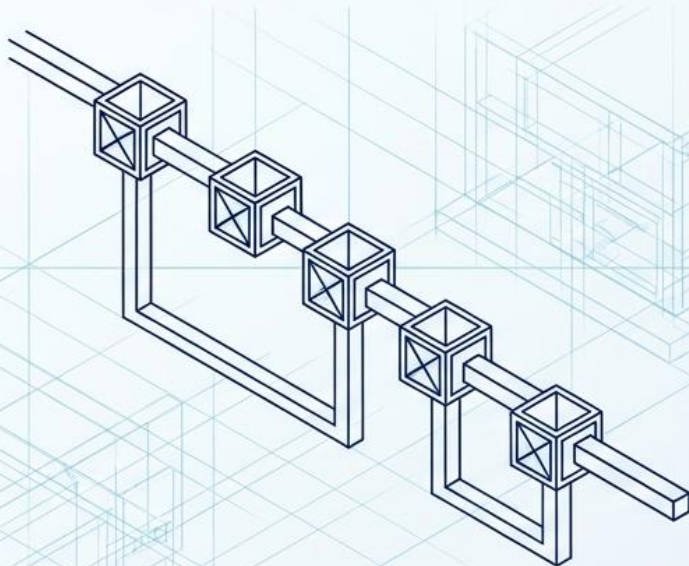
Output



https://quantumai-lively.app.n8n.cloud/workflow/KVNFE6Yxfm9s9uCn?projectId=qWoC5wFQ5qcl8WnB&uiContext=workflow_list

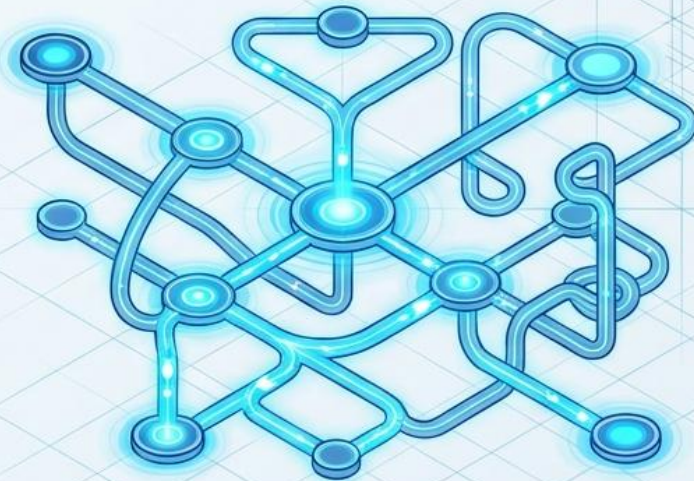
Moving Beyond Automation

Traditional Automation



Traditional software follows a recipe: rigid, predefined logic executing a fixed set of instructions.

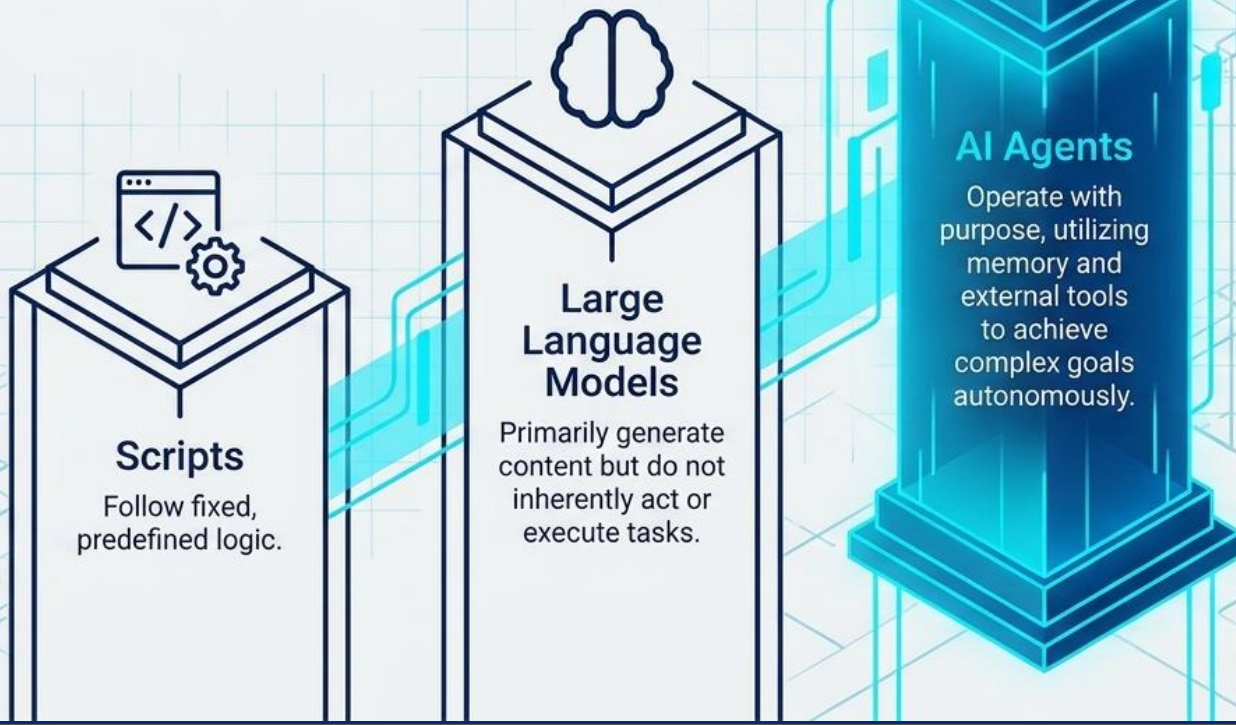
Agentic Autonomy



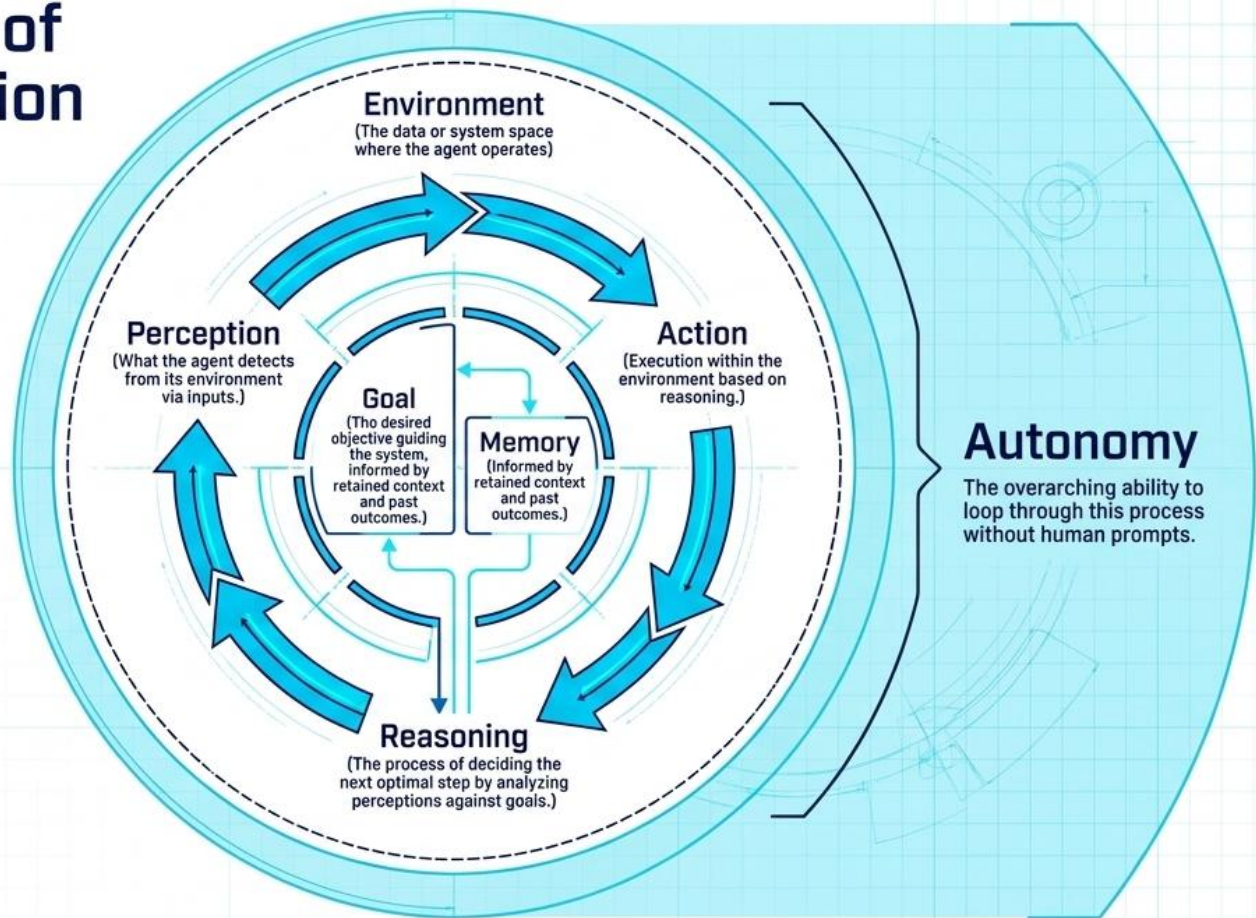
Intelligent AI agents are master chefs: they autonomously reason, plan, and adapt their actions to navigate complex, unpredictable scenarios.

Defining the AI Agent

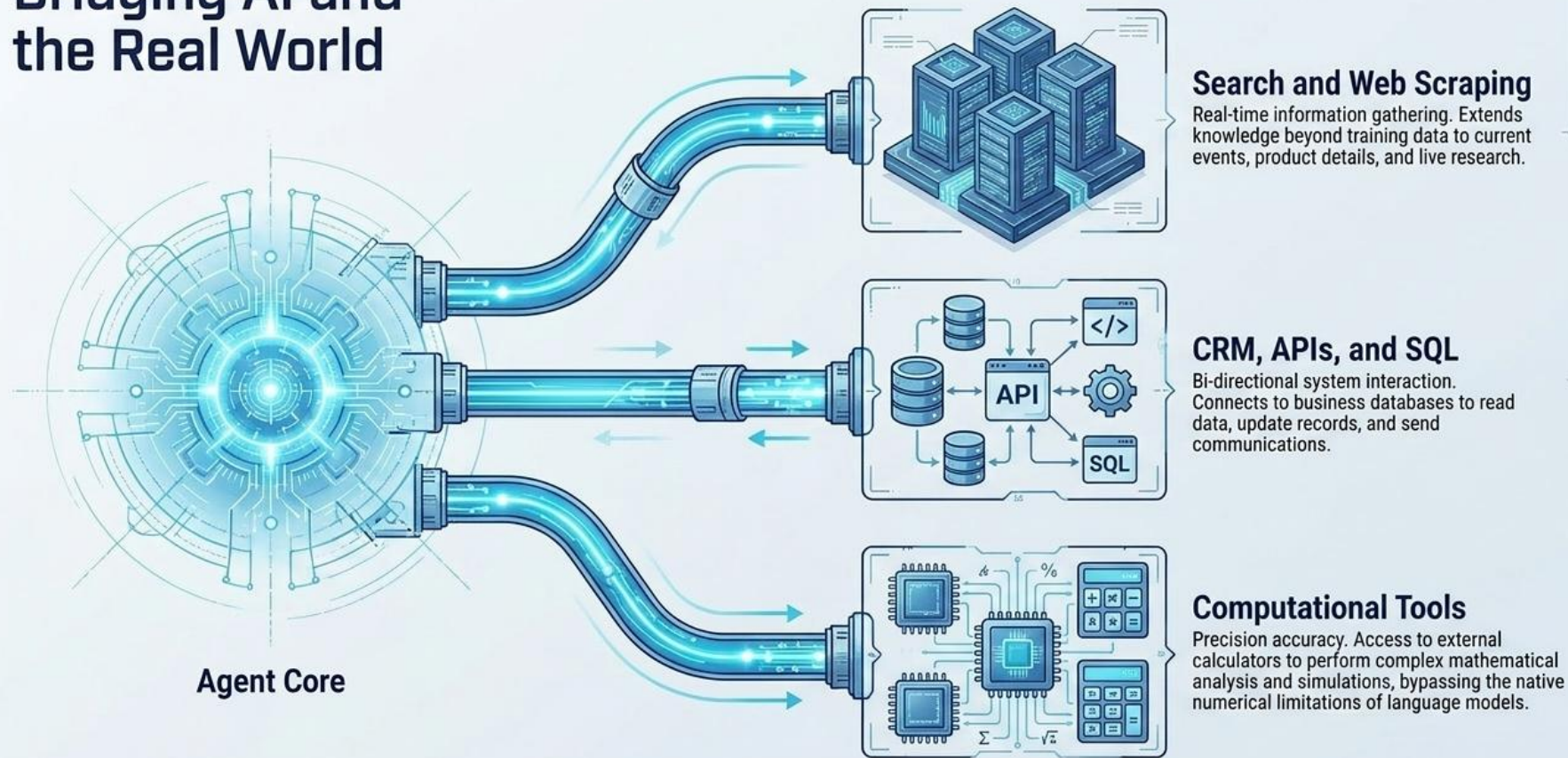
An AI agent is a system that can perceive, reason, and act toward a goal with autonomy.



The Anatomy of Digital Cognition



Bridging AI and the Real World



Advanced Workflows: The ReAct Pattern

Reason

The agent analyzes its current context, perceptions, and overall goal to plan the optimal next step or tool selection.

Observe & Adapt

The agent observes the real-world outcome of the action, feeding the result back into the next reasoning phase until the task is complete.

Act

The agent executes the chosen action—searching a database, calling an API, or formulating a response.

Applied Use Cases: Customer Support triaging, Automated Data Entry.

The Agentic Workforce: Applied Business Utility



Customer Support Agent

Identity: Digital first responder for rapid problem resolution.

Purpose: Reduce **ticket volume** and enhance **satisfaction**.

Function: Troubleshoot issues, guide users through workflows, intelligently escalate edge cases.



Analyst Agent

Identity: Data-driven researcher and summarizer.

Purpose: Accelerate **competitive analysis** and **data synthesis**.

Function: Gather multi-source data, scan market trends, generate executive reports.



Sales Assistant

Identity: Digital partner for lead nurturing.

Purpose: Automate **initial qualification** and provide **instant responses**.

Function: Qualify leads, schedule follow-ups, and manage CRM updates autonomously.

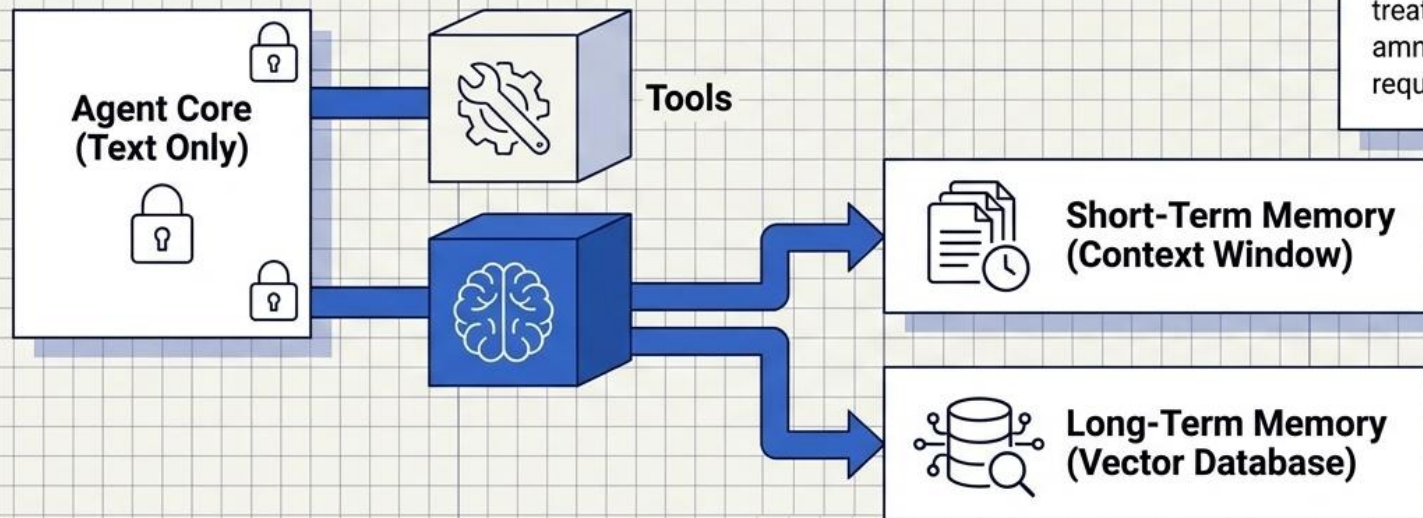


The Strategic Mandate

- AI agents signify a permanent shift from rigid, scripted software to dynamic, goal-oriented digital co-pilots.
- By combining contextual memory, multi-step reasoning, and external tool utilization, agents do not just execute tasks—they autonomously solve complex business problems.
- The future belongs to organizations that integrate these autonomous systems to augment human strategy, scale personalized interactions, and transform operational speed.

Memory: The Brain of the Agent

Memory enables an agent to retain and reuse information across interactions. Without it, an agent has zero continuity; with it, the agent builds an evolving understanding of the user and the task.



Core Architectural Principle: The Continuity Rule

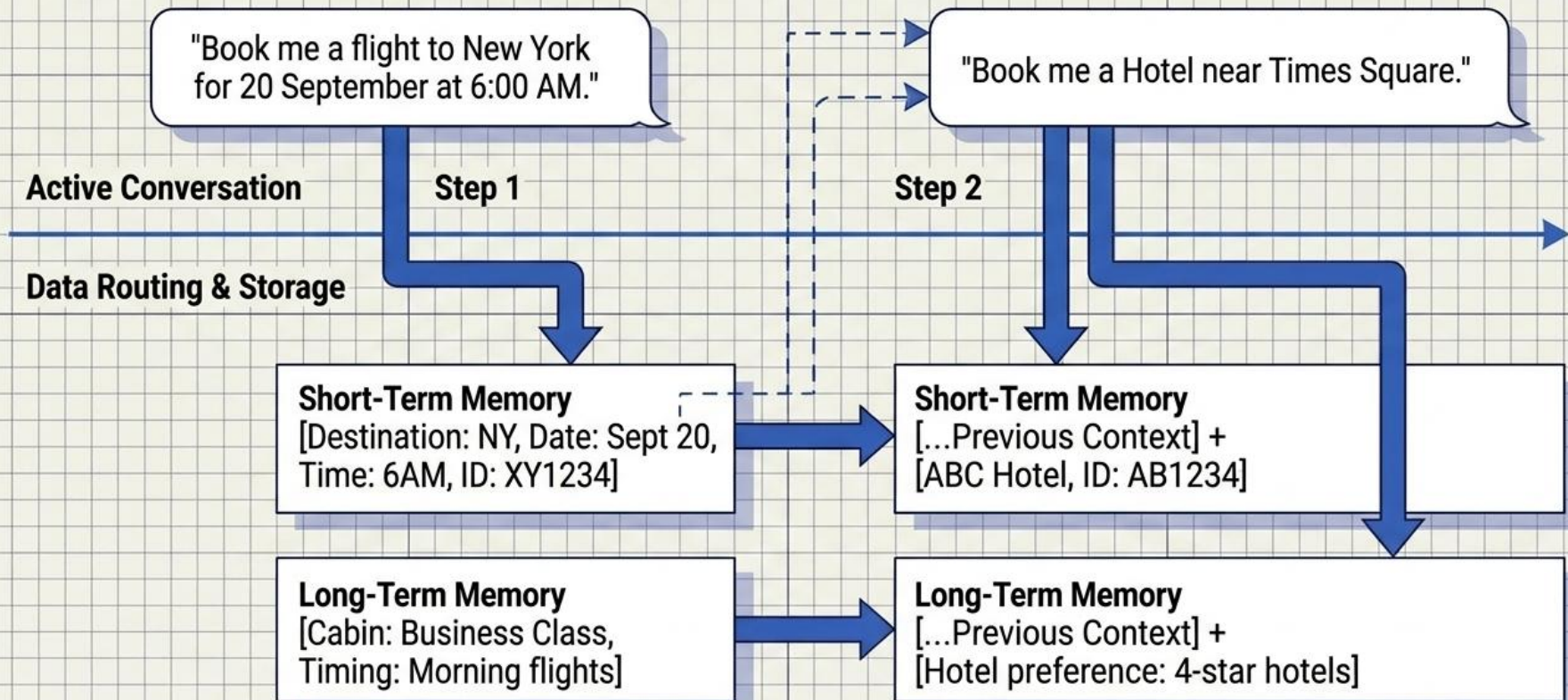
The primary benefit of adding memory to agents is providing personalization and context continuity, preventing the system from treating every query as an amnesiac, standalone request.

The Memory Matrix:

Short-Term vs. Long-Term

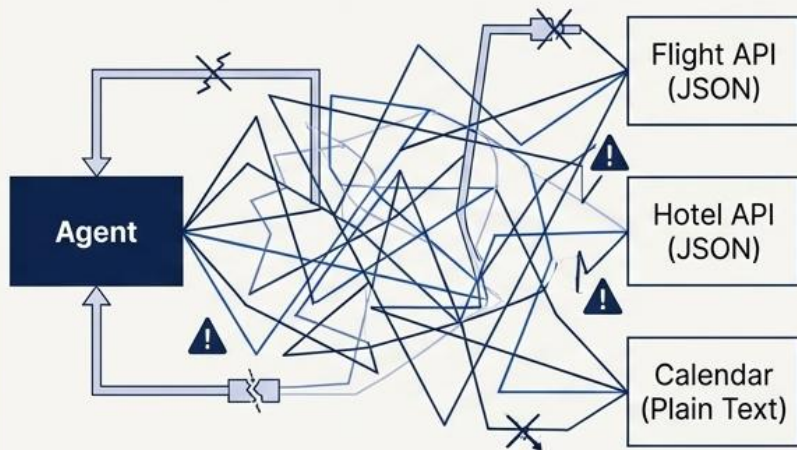
	Short-Term Memory	Long-Term Memory
Scope & Duration	Used within a single, ongoing conversation to track recent inputs, instructions, and intermediate reasoning.	Spans multiple disparate conversations to retain historical interactions, user preferences, and vital information.
Storage Location	Held temporarily in the conversation context or the immediate LLM prompt window.	Stored persistently in external architectures like databases, vector stores, or dedicated memory services.
Primary Function	Ensures highly coherent, context-aware responses without losing the thread of the current interaction.	Enables compounding personalization, cross-session consistency, and a drastically improved user experience over time.

Continuous Context in Action



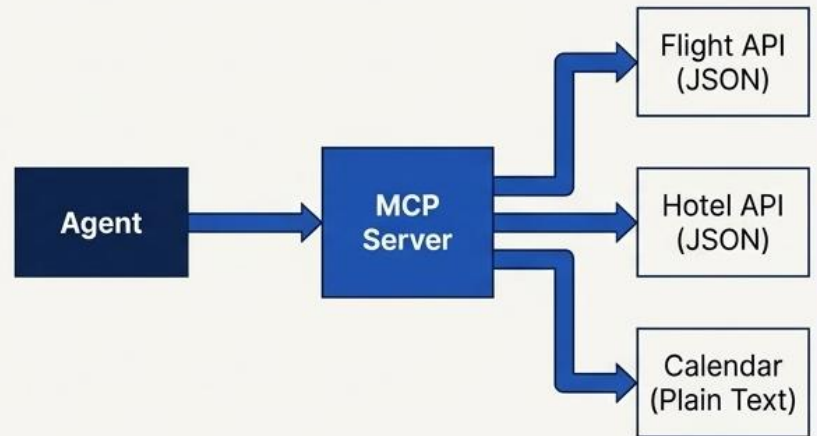
The Model Context Protocol (MCP)

Before: Integration Spaghetti



The Problem: Different external tools demand varied input/output formats. This tight coupling causes inconsistencies, errors, and scaling bottlenecks.

After: The MCP Hub



The Solution: MCP solves integration chaos by defining an open, common protocol for how agents discover and access external capabilities at runtime.

The MCP Mandate: Standardize or Stagnate. Without a unified protocol, agents lose crucial context, leading to mistakes, increased complexity, and unreliable decisions.

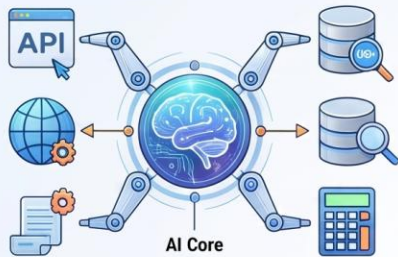
Anatomy of an AI Agent: Tools, Memory, and Connectivity

How Core Components Transform an LLM into a Functional AI Agent

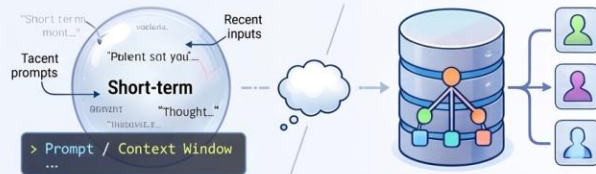
The Pillars of Agent Capability

Tools: Beyond Text Generation

External systems (APIs, SQL, Calculators) enabling agents to interact with real-world data.



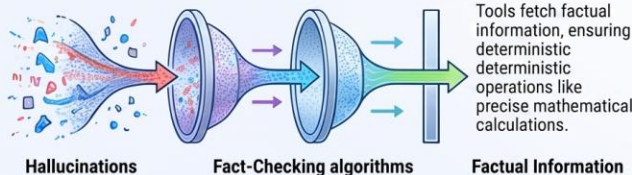
Short-term vs. Long-term Memory



Prompt / Context Window
Tracks reasoning and recent inputs

Vector Stores / Databases
Retains history and personalization

Reducing Hallucinations via Data



Model Context Protocol (MCP)

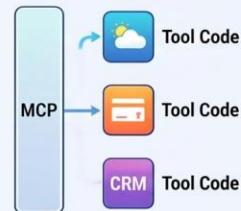
The Standard for Connectivity

An open standard enabling agents to discover and use tools in a unified way.



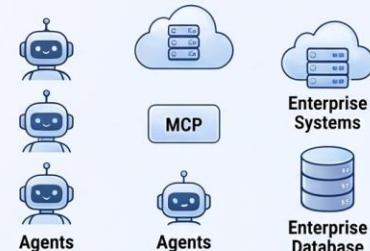
Decoupling Logic from Implementation

MCP separates agent reasoning from specific tool code, making integrations reusable and maintainable.



Enterprise-Ready Architecture

Solves "tight coupling" issues to create scalable and consistent AI system designs.



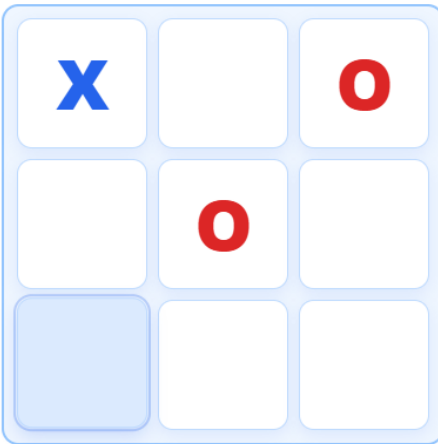
Agentic Tic Tac Toe

Answer correctly to claim a square. An incorrect response passes the turn and leaves the square open.

AI AGENTS • AGENTIC WORKFLOWS • MEMORY • TOOLS • MCP • REACT • REFLECTION

Player X's turn

X



Player O claimed the square. Player X is next.

KNOWLEDGE CHALLENGE

What is the agent's environment?

- ☐ A A list of possible final answers
- ☐ B The model's training hardware
- ☐ C Only the text of the system prompt
- ☐ D The data or system space in which the agent operates and interacts

Show Hint

Restart Game

https://huggingface.co/spaces/eaglelandsonce/Agentic_TicTacToe

Case Study - Reimbursement Automation



Business Context



- In many mid-sized and large organizations, **employee expense processing and reimbursement** continues to rely heavily on manual, paper-based workflows. Employees submit physical or scanned receipts, which finance teams must individually review, categorize, and validate against company expense policies. This process demands significant time and effort to verify expense details such as category, amount, date, vendor, and compliance requirements.
- As organizations grow, the **volume of expense claims increases substantially**, placing additional strain on finance teams. Manual processing creates operational bottlenecks, slows reimbursement cycles, and raises processing costs. It also increases the likelihood of human errors, duplicate reimbursements, and policy violations. Delays in reimbursements can negatively affect employee satisfaction and erode confidence in internal financial processes.
- Additionally, expense policy interpretation often varies across reviewers, leading to **inconsistent approval decisions** and increased audit and compliance risks. The lack of automation limits real-time visibility into organizational spending patterns, making it difficult for finance teams to proactively enforce controls, detect anomalies, or identify potential misuse early in the process.



Objective

The objective is to simplify and streamline the expense processing and reimbursement workflow by leveraging intelligent automation. Routine tasks such as receipt data extraction, expense categorization, and policy validation can be handled automatically, reducing manual effort and processing time. This enables finance teams to focus on exception handling, compliance oversight, and financial analysis rather than repetitive verification tasks.

- Reduce manual effort in reviewing and processing expense claims
- Minimize errors and policy non-compliance in reimbursements
- Accelerate reimbursement turnaround time for employees



Solution Approach

Workflow 1: Reimbursement Agent (Main Orchestrator)

- Receives employee bill via chat trigger and classifies raw input into **emp_id**, **emp_name**, and **raw_bill** using GPT-4
- Passes structured context to an AI Agent
- Agent calls MCP tools in sequence: structure bill → get caps → get utilization → calculate → send email
- Returns a final JSON summary with approved/rejected amounts per category

Workflow 2: MCP Server (Tool Registry)

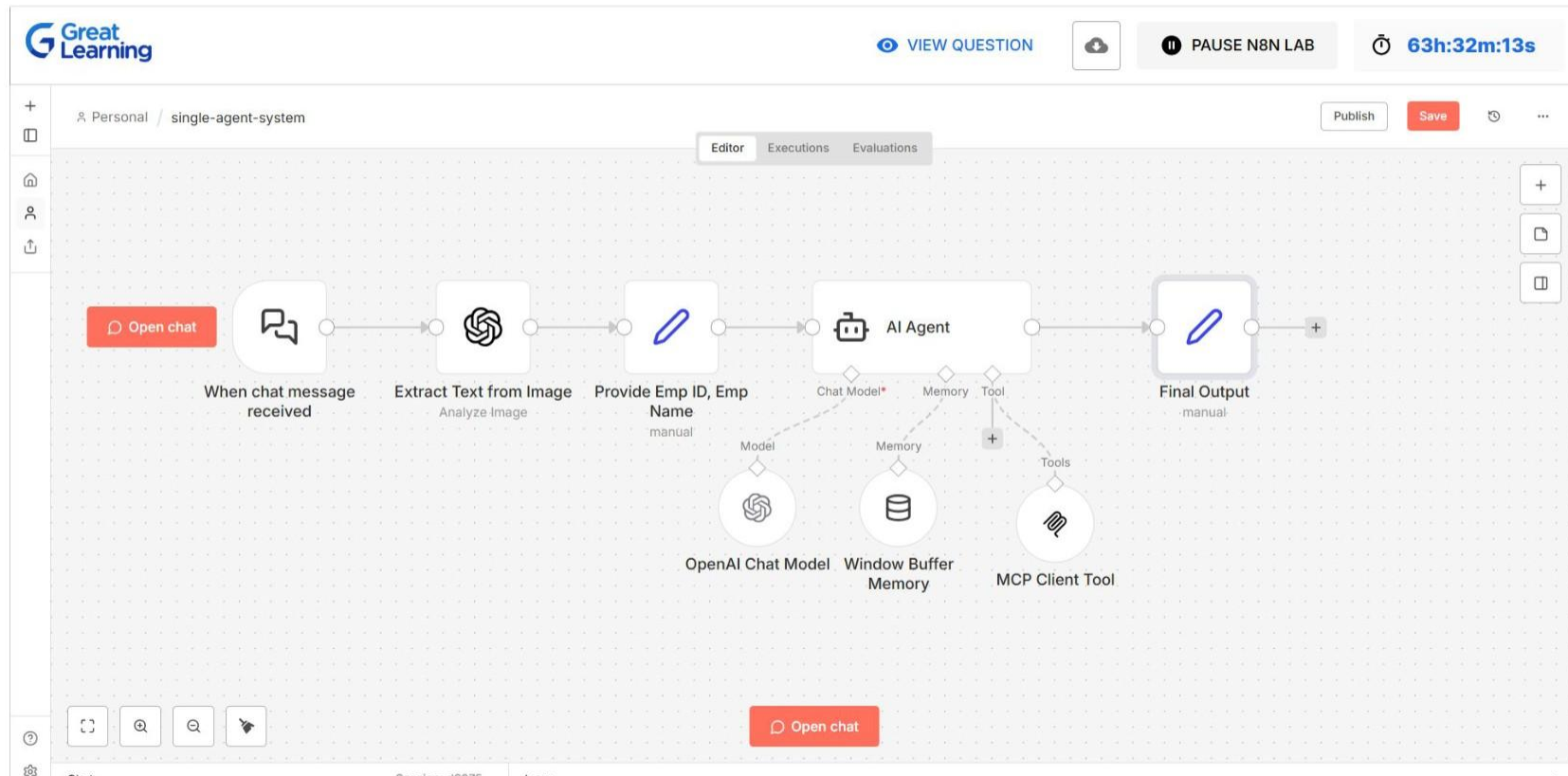
- Exposes four tools over SSE to the agent:
 - **structure_bill** - delegates to Sub-workflow for bill parsing
 - **get_reimbursement_caps** - queries SQLite for per-category caps by employee position
 - **get_past_utilization** - queries SQLite for already-claimed amounts this period
 - **send_email** - sends the reimbursement decision email via SMTP



Workflow 3: Sub-Workflow - Structure Bill

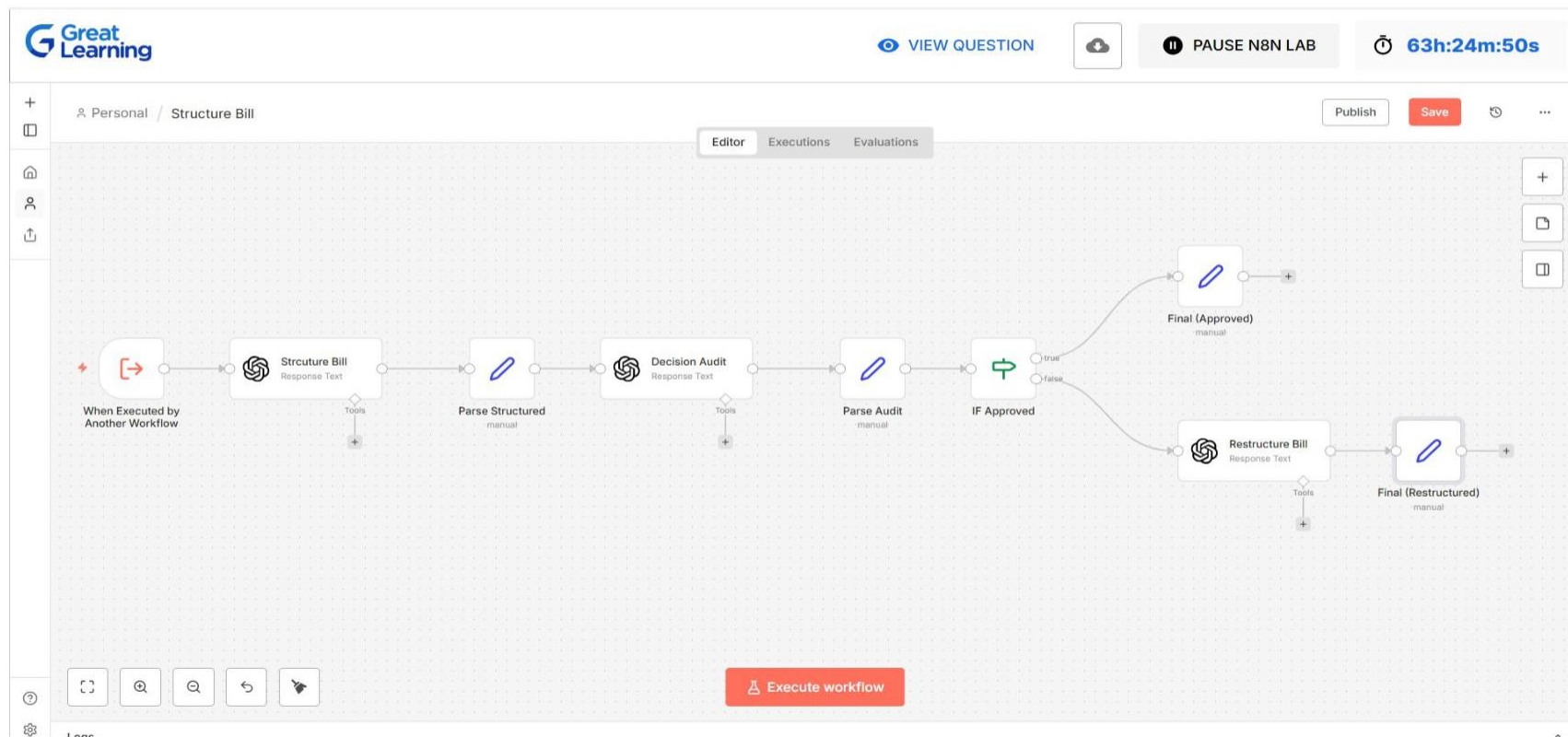
- Accepts raw bill text from the MCP server trigger
- **Structure** → GPT-4.1 extracts category-wise tax-inclusive totals (food, travel, hotel, others)
- **Audit** → A second GPT-4.1 call verifies arithmetic and categorization integrity
- If audit passes → returns structured bill as-is
- If audit fails → a third GPT-4.1 call corrects only the flagged fields, then returns the fixed output

Solution Workflow - Single Agent Workflow



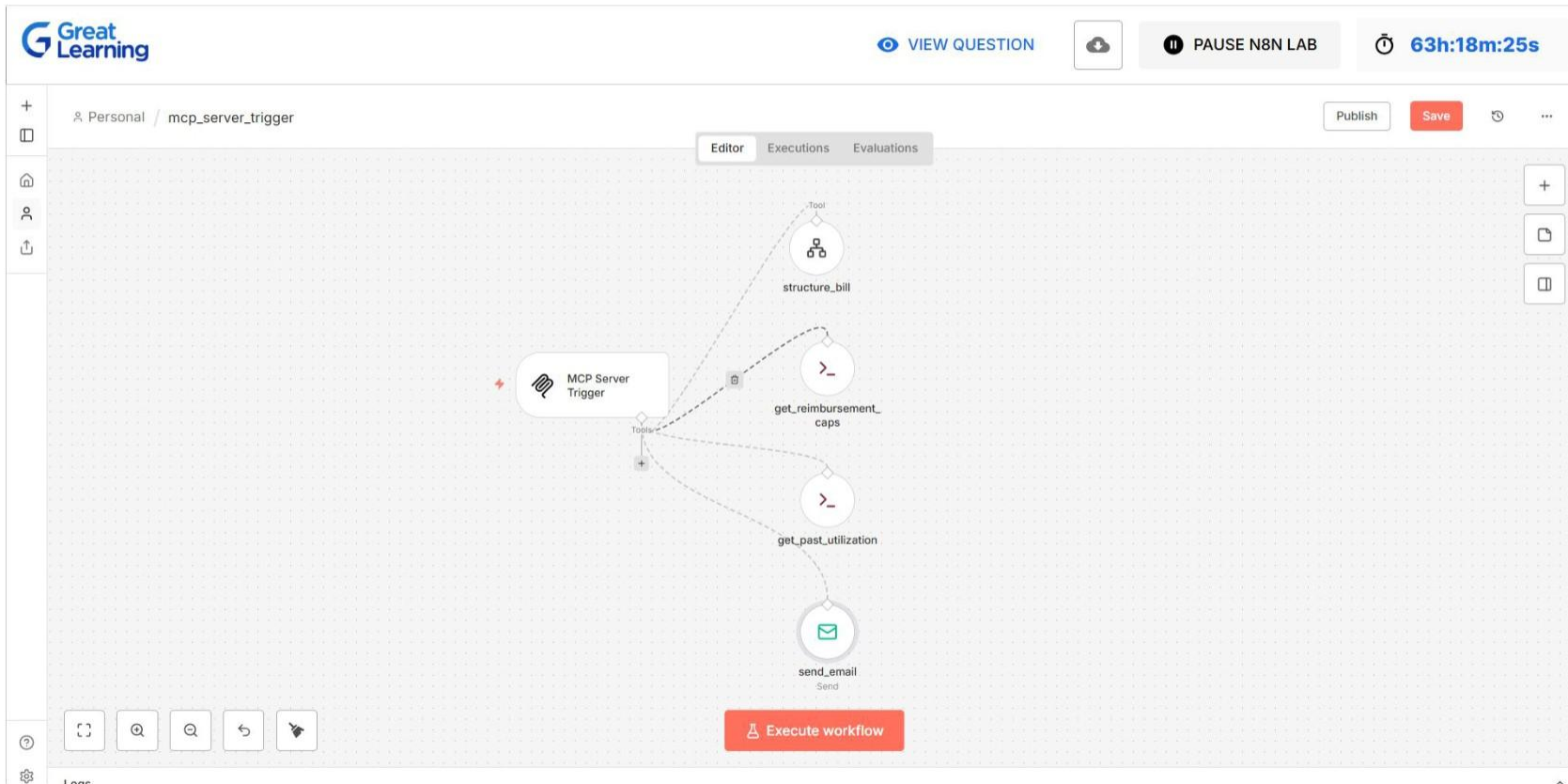


Solution Workflow - Sub Workflow for Bill Structuring





Solution Workflow - MCP Server Trigger Workflow



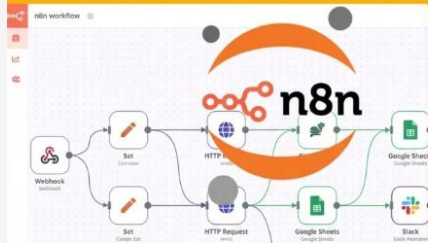
Login to n8n



Welcome to the n8n Lab

- ✓ **Visual Workflow Building**
Design, test, and orchestrate node-based automations interactively using the n8n lab
- ✓ **Save and Resume**
Your workflows are saved and can be resumed or iterated on later
- ✓ **Export**
Export workflows for deployment or future reuse

RESUME N8N LAB



https://olympus.mygreatlearning.com/gl_coding_labs/aB3r93yA

Data Set



receipts



reimbursement



Workflows



✓ e001_Amit_Sharm
ma



✓ e002_Neha_Verma



✓ e003_Rahul_Mehta



MCP Server Trigger

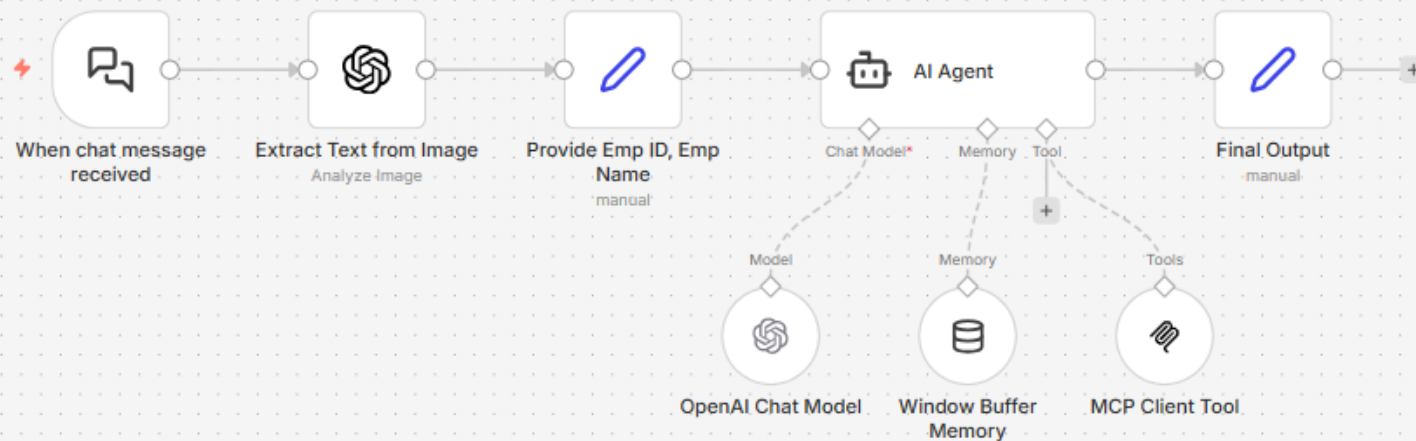


Reimbursement Agent



Sub_Structure Bill

https://olympus.mygreatlearning.com/gl_coding_labs/aB3r93yA



Use Published Address of MCP Server
Publish your Structure Bill (select it)!



Step 2.11 - Save and Publish the Sub Workflow

Great Learning

VIEW QUESTION

PAUSE N8N LAB 63h:24m:50s

Personal / Structure Bill

Editor Executions Evaluations

When Executed by Another Workflow

Structure Bill Response Text

Parse Structured manual

Decision Audit Response Text

Parse Audit manual

IF Approved

Final (Approved) manual

Restructure Bill Response Text

Final (Restructured) manual

Execute workflow

Once workflow prepared with all the required Nodes, Click on Save and Publish the Workflow

Publish Save



Step 3.3 - Add Call n8n Workflow Tool

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:21m:49s

structure_bill

INPUT Mapping From AI

? Parent nodes

No input connected

Parameters Settings Execute step

See an example of a workflow to suggest meeting slots using AI [here](#).

Description

Convert raw bill text into a clean structured JSON with category-wise (food/travel/hotel/others) tax-inclusive totals. Internally runs a structuring agent, an arithmetic

This tool will call the workflow you define below, and look in the last node for the response. The workflow needs to start with an Execute Workflow trigger

Source

Database

Workflow

From list Sub: Structure Bill

Workflow Inputs

raw_bill

fx {{ \$fromAI('raw_bill', '', 'string') }}

Add workflow input to send

I wish this node would...

OUTPUT

Click on the **Dropdown** list and select the published sub workflow

Note: If sub workflow not published, we can't see the sub workflow option in this list

No output data

Output will appear here once the parent node is run



Step 3.7 - Save and Publish the Workflow

VIEW QUESTION

PAUSE N8N LAB

63h:18m:25s

Personal / mcp_server_trigger

Editor

Executions

Evaluations

```
graph LR; Trigger[MCP Server Trigger] --> structure_bill[structure_bill]; Trigger --> get_reimbursement_caps[get_reimbursement_caps]; Trigger --> get_past_utilization[get_past_utilization]; Trigger --> send_email[send_email];
```

Publish

Save

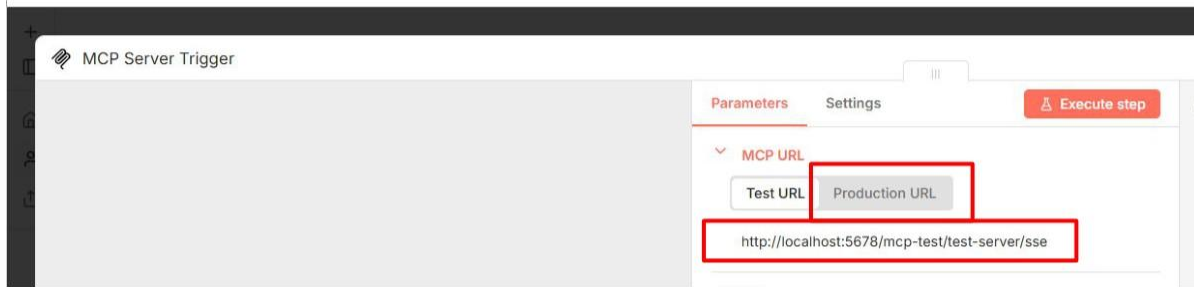
Once workflow prepared with all the required Nodes, Click on Save and Publish the Workflow

Execute workflow

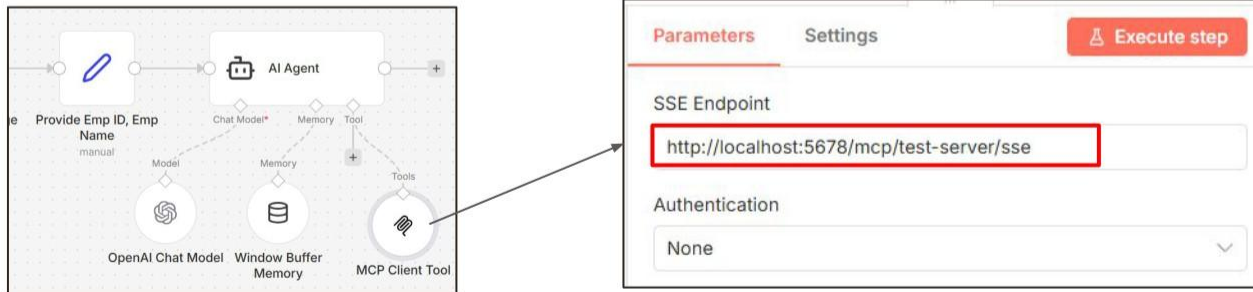


General Guidelines (Must Follow)

- Navigate to **Sub Workflow** - Open the **MCP Server Trigger** - Copy the **Production URL**, not Test URL



- Navigate to **Single Agent Workflow** - Open the **MCP Client Tool** - Paste the **Production URL** in the SSE Endpoint



- Upload the [reimbursement.db](#) database provided in to n8n lab before executing the workflows

Workflow - 1: Single Agent Workflow

Step 1.1 - Create a New Workflow



[VIEW QUESTION](#)



 PAUSE N8N LAB

 63h:37m:36s

+

□

🏠

👤

📶

Personal / single-agent-system

Publish

Saved

🕒

⋮

Editor

Executions

Evaluations

+

📄

📱

🔄

🔍

🔍

🔧

+

Add first step...



Step 1.2 - Add Chat Trigger

VIEW QUESTION

PAUSE N8N LAB

63h:37m:10s

Personal / single-agent-system

Publish

Saved

...

Editor

Executions

Evaluations

+

Add first step...

What triggers this workflow?
A trigger is a step that starts your workflow.

chat

Chat Trigger

AI Agent

OpenAI

Chat Memory Manager

Chat Data
Basic Chat Data Node

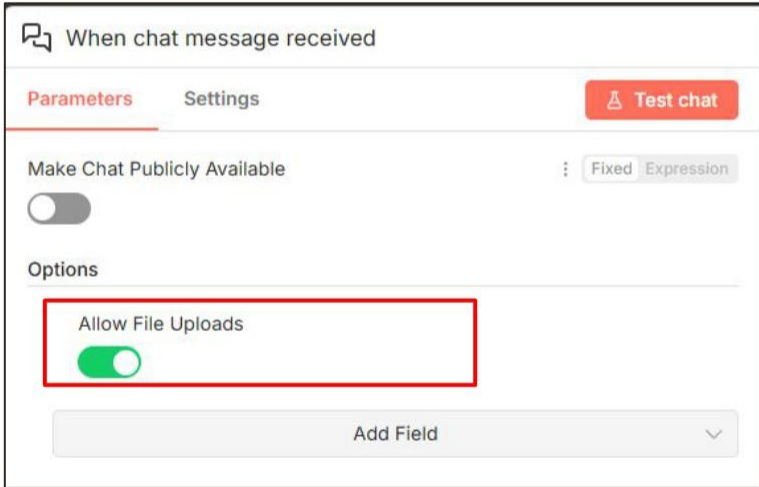
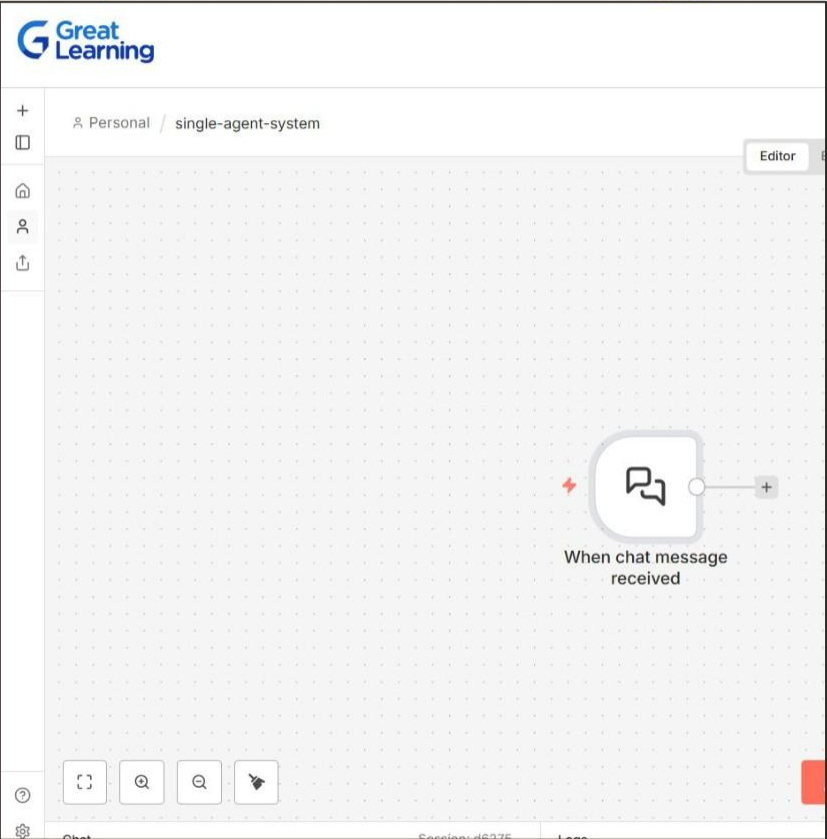
2Chat
Sends data to 2Chat

Channel Talk
Starts the workflow when Channel Talk events occur

Groq Chat Model

QuickChat

Step 1.2 - Add Chat Trigger



Step 1.3 - Add Open AI Image Model



[VIEW QUESTION](#)



[PAUSE N8N LAB](#)

[63h:36m:33s](#)

+

□

🏠

👤

📁

Personal / single-agent-system

Editor

Executions

Evaluations



When chat message received

🔄

🔍

🔍

📌

[Open chat](#)

OpenAI

Actions (16)

TEXT ACTIONS

 Message a model

 Classify text for violations

IMAGE ACTIONS

 Analyze image

 Generate an image

 Edit image

AUDIO ACTIONS

 Generate audio

 Transcribe a recording

 Translate a recording

FILE ACTIONS

 Delete a file

Step 1.3 - Add Open AI Image Model



[VIEW QUESTION](#)



[PAUSE N8N LAB](#)

[63h:36m:14s](#)

 Extract Text from Image

INPUT

→|

No input data

Execute previous nodes
to view input data

Parameters Settings [Execute step](#)

Credential to connect with

OpenAI account

Resource

Image

Operation

Analyze Image

Model

From list GPT-4O

Text Input

Extract the text details from the image uploaded, make sure you extract everything

Input Type

Binary File(s)

Input Data Field Name

data0

The name of the input field containing the binary file data to be processed

Simplify Output

☒

OUTPUT

Docs



|→

No output data

Execute step
or set mock data

Step 1.4 - Add Edit Fields(Set)

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:36m:00s

Personal / single-agent-system

Editor Executions Evaluations

When chat message received

Extract Text from Image
Analyze Image

+

What happens next?

edit

Edit Image
Edits an image like blur, resize or adding border and text

Edit Fields (Set)
Modify, add, or remove item fields

Reddit

PredictLeads
Get company insights from PredictLeads' official node

Remove Duplicates
Delete items with matching field values

Bedrijfsdata
Get data from the Bedrijfsdata API.

Read/Write Files from Disk
Read or write files from the computer that runs n8n

Ldap

Open chat



Step 1.4 - Add Edit Fields(Set)

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:35m:39s

Provide Emp ID, Emp Name

INPUT

No input data
Execute previous nodes
to view input data

ParametersSettingsExecute step

Mode
Manual Mapping

Fields to Set

Emp_ID
String
E002
E002

emp_name
String
Neha Verma
Neha Verma

raw_bill
String
{{ \$json['0'].content[0].text }}

receiver_email
String
praneeth78211@gmail.com

OUTPUT

No output data
Execute step
or set mock data

Step 1.5 - Add AI Agent



VIEW QUESTION



PAUSE N8N LAB

63h:35m:24s

Personal / single-agent-system

Editor | Executions | Evaluations



```
graph LR; A[When chat message received] --> B[Extract Text from Image]; B --> C[Provide Emp ID, Emp Name]; C --> D[+];
```

When chat message received

Extract Text from Image

Provide Emp ID, Emp Name

+

What happens next?

aiag

AI Agent

Mallabe Images

ActiveCampaign

AWS Certificate Manager

Picsart Image AI

WAWCD Plugin Send Message

WAWCD Plugin Receive Message

Open chat



Step 1.5 - Add AI Agent



[VIEW QUESTION](#)



[PAUSE N8N LAB](#)

[63h:34m:49s](#)

AI Agent

INPUT

→|

No input data

Execute previous nodes
to view input data

Parameters Settings **Execute step**

Source for Prompt (User Message)

Define below

Prompt (User Message)

fx

Process this employee bill:
- Emp_ID: {{ \$json.Emp_ID }}
- emp_name: {{ \$json.emp_name }}
- receiver_email: {{ \$json.receiver_email }}
- raw_bill:
{{ \$json.raw_bill }}

Require Specific Output Format

☐

Options

System Message

fx

You are a reimbursement-processing assistant. You receive an employee bill with:
- Emp_ID
- emp_name
- raw_bill

You are a reimbursement-processing assistant. You...

Add Option

Chat Model * Memory Tool

OUTPUT

|→

No output data

Execute step
or set mock data



Step 1.6 - Add OpenAI Chat Model



VIEW QUESTION



PAUSE N8N LAB

63h:34m:24s

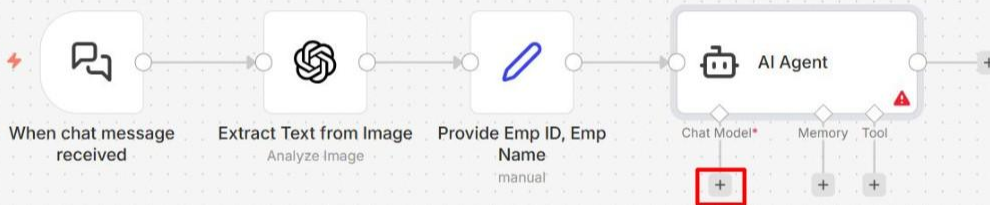
Personal / single-agent-system

Publish

Save



Editor Executions Evaluations



Language Models

cha

Groq Chat Model
Language Model Groq

OpenAI Chat Model
For advanced usage with an AI chain

Ollama Chat Model
Language Model Ollama

DeepSeek Chat Model
For advanced usage with an AI chain

xAI Grok Chat Model
For advanced usage with an AI chain

Lemonade Chat Model
Language Model Lemonade Chat

Anthropic Chat Model
Language Model Anthropic

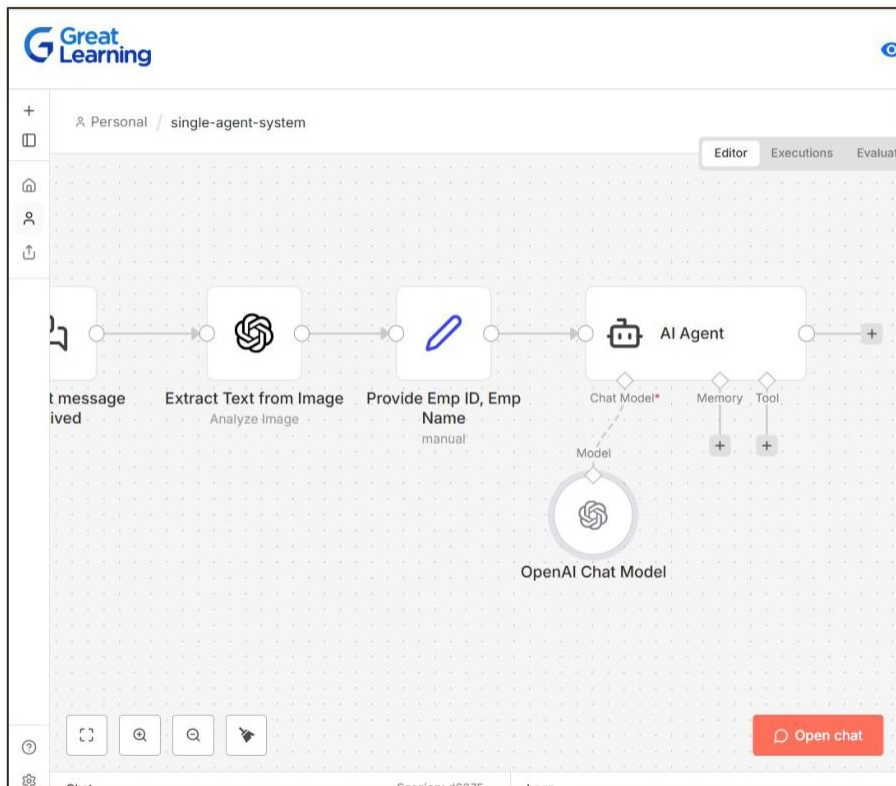
Google Gemini Chat Model
Chat Model Google Gemini

Vertex AI Gemini Chat Model

Open chat



Step 1.6 - Add OpenAI Chat Model



Parameters Settings

Credential to connect with
OpenAi account

Model
From list gpt-4o-mini

Options
No properties
Add Option



Step 1.7 - Add Simple Memory

VIEW QUESTION PAUSE N8N LAB 63h:33m:47s

Personal / single-agent-system

Editor Executions Evaluations

AI Agent

Chat Model* Memory Tool

OpenAI Chat Model

Open chat

Memory

Memory allows an AI model to remember and reference past interactions with it

For beginners

- Simple Memory**
Stores in n8n memory, so no credentials required

Other memories

- MongoDB Chat Memory**
Stores the chat history in MongoDB collection.
- Motorhead**
Use Motorhead Memory
- Postgres Chat Memory**
Stores the chat history in Postgres table.
- Redis Chat Memory**
Stores the chat history in Redis.
- Xata**
Use Xata Memory



Define below

session2

10

How many past interactions the model receives as context

Great Learning

VIEW QUESTION



PAUSE N8N LAB

63h:33m:14s

Personal / single-agent-system

Editor

Executions

Evaluations

Tools

mcp

Model Context Protocol

MCP Client Tool
Connect tools from an MCP Server

Other Tools

Mailchimp Tool
Consume Mailchimp API

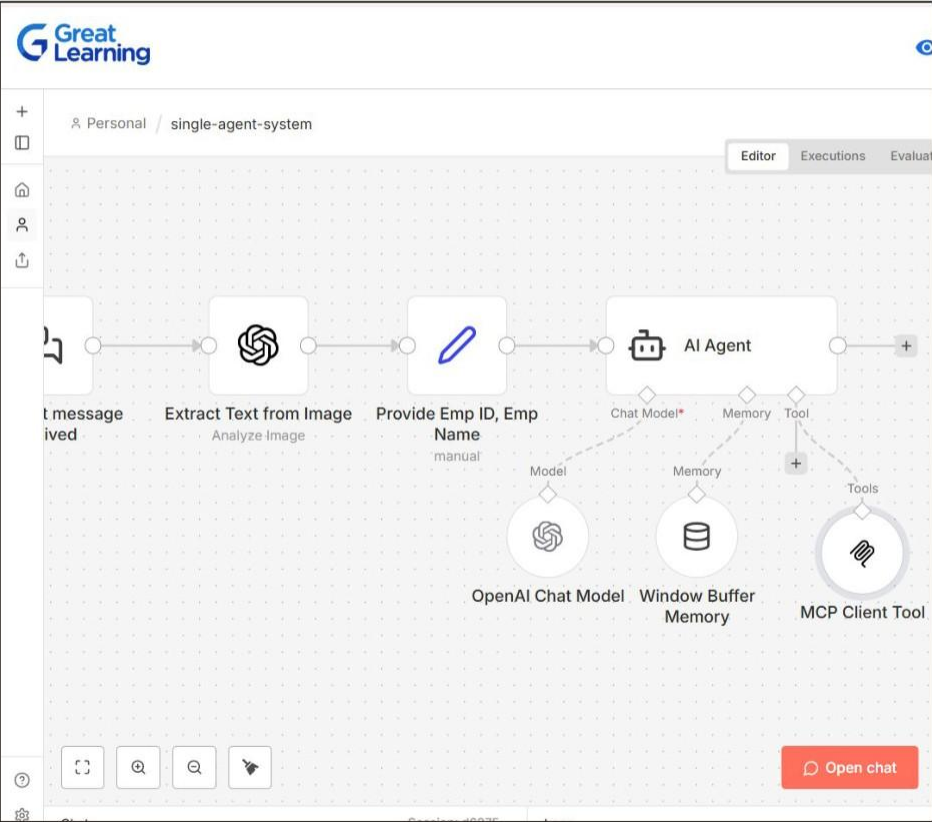
Microsoft SharePoint Tool
Interact with Microsoft SharePoint API

Microsoft Graph Security Tool
Consume the Microsoft Graph Security API

Open chat

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Step 1.8 - Add MCP Client Tool



Parameters

Settings

Execute step

SSE Endpoint

http://localhost:5678/mcp/test-server/sse

Authentication

None

Tools to Include

All

Options

Timeout

60000



Step 1.9 - Add Edit Fields(Set)

VIEW QUESTION

PAUSE N8N LAB

63h:32m:43s

Personal / single-agent-system

Editor

Executions

Evaluations

What happens next?

ed

Edit Image
Edits an image like blur, resize or adding border and text

Redis

Edit Fields (Set)
Modify, add, or remove item fields

Medium

LinkedIn

Convert to File
Convert JSON data to binary data

Reddit

Extract from File
Convert binary data to JSON

PredictLeads
Get company insights from PredictLeads

Open chat



Step 1.9 - Add Edit Fields(Set)



[VIEW QUESTION](#)



[PAUSE N8N LAB](#)

[63h:32m:24s](#)

Final Output

INPUT

No input data

[Execute previous nodes](#)
to view input data

Parameters Settings [Execute step](#)

Mode

Manual Mapping

Fields to Set

Emp_ID

T String

= {{ \$('Provide Emp ID, Emp Name').item.json.Emp_ID }}

emp_name

T String

= {{ \$('Provide Emp ID, Emp Name').item.json.emp_name }}

result

T String

= {{ \$json.output }}

Drag input fields here or [Add Field](#)

Include Other Input Fields

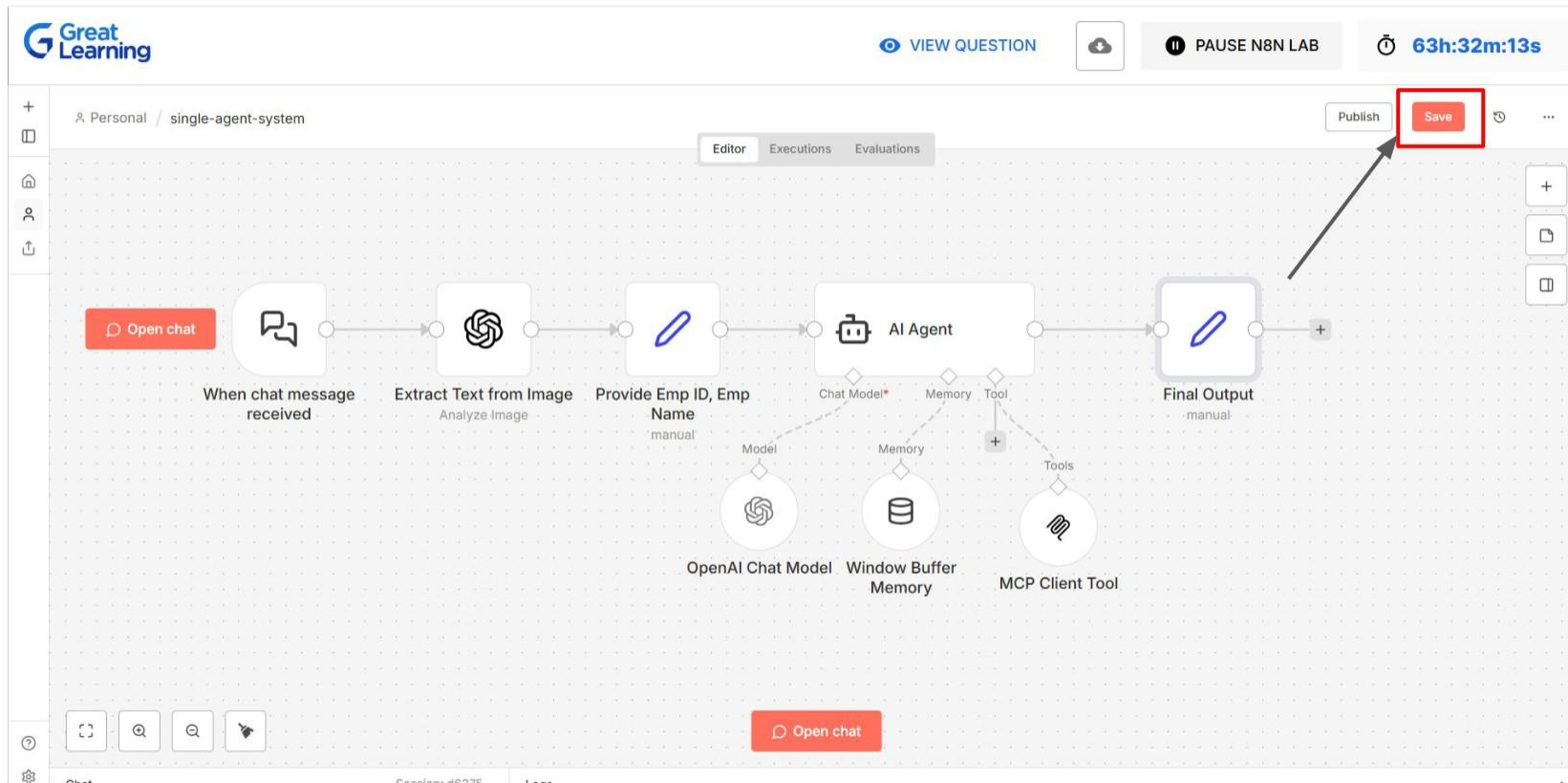
OUTPUT

No output data

[Execute step](#)
or [set mock data](#)



Step 1.10 - Final Workflow



Workflow - 2: Sub Workflow



Step 2.1 - Create a New Workflow

VIEW QUESTION

PAUSE N8N LAB

63h:30m:44s

Personal / Structure Bill

Publish

Saved

...

Editor

Executions

Evaluations

Add first step...

Log



 63h:30m:30s

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Step 2.2 - Add “When executed by another workflow”

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:30m:21s

When Executed by Another Workflow

Parameters Settings Execute step

Input data mode
Define using JSON example

Provide an example object to infer fields and their types. To allow any type for a given field, set the value to null.

JSON Example

```
1 {
2   "raw_bill": "text"
3 }
```

OUTPUT

No trigger output
Test this trigger
or set mock data

Step 2.3 - Add OpenAI(Message a model)



[VIEW QUESTION](#)



[PAUSE N8N LAB](#)

[63h:29m:50s](#)

Personal / Structure Bill

Editor Executions Evaluations

[Publish](#)

[Save](#)



OpenAI

Search OpenAI Actions...

Actions (16)

TEXT ACTIONS

Message a model

Classify text for violations

IMAGE ACTIONS

Analyze image

Generate an image

Edit image

AUDIO ACTIONS

Generate audio

Transcribe a recording

Translate a recording

FILE ACTIONS

Delete a file



Execute workflow



Step 2.3 - Add OpenAI(Message a model)

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:29m:29s

Strcuture Bill

INPUT

→|
No input data
Execute previous nodes
to view input data

Parameters

Settings

Execute step

Credential to connect with
OpenAi account

Resource
Text

Operation
Message a Model

Model
From list GPT-4.1

Messages
Type
Text
Role
System
Prompt
ROLE
You are a senior Corporate expense anal
yst specializing in vendor-agnostic exp
ense normalization across hotels, resta
Tools

OUTPUT

|→
No output data
Execute step
or set mock data



Step 2.4 - Add Edit Fields(Set)



VIEW QUESTION



PAUSE N8N LAB

63h:29m:03s

Personal / Structure Bill

Publish

Save



Editor

Executions

Evaluations

What happens next?



ed



Edit Image



Edits an image like blur, resize or adding border and text



Redis



Edit Fields (Set)



Modify, add, or remove item fields



Reddit



Medium



Execution Data



Add execution data for search



Remove Duplicates



Delete items with matching field values



Convert to File



Convert JSON data to binary data



Hedy



Execute workflow



Step 2.4 - Add Edit Fields(Set)

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:28m:49s

Parse Structured

INPUT

→|

No input data

Execute previous nodes
to view input data

Parameters Settings Execute step

Mode

Manual Mapping

Fields to Set

raw_bill

T String

= `={{ $('When Executed by Another Workflow').item.json.raw_bill }}`

structured_bill

T String

= `{{ JSON.stringify($json.output) }}`

Drag input fields here or Add field

Include Other Input Fields

☐

Options

No properties

Add option

OUTPUT

|→

No output data

Execute step
or set mock data



Step 2.5 - Add OpenAI(Message a model)



VIEW QUESTION



PAUSE N8N LAB

63h:28m:32s



Personal / Structure Bill

Editor

Executions

Evaluations

Publish

Save



OpenAI

Search OpenAI Actions...

Actions (16)

TEXT ACTIONS

Message a model

Classify text for violations

IMAGE ACTIONS

Analyze image

Generate an image

Edit image

AUDIO ACTIONS

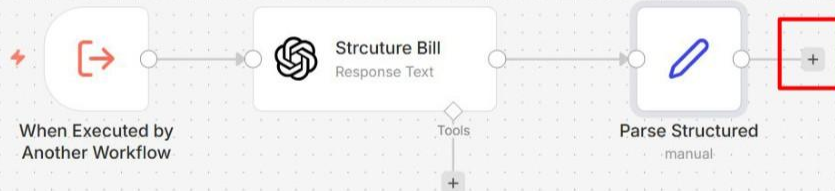
Generate audio

Transcribe a recording

Translate a recording

FILE ACTIONS

Delete a file



Execute workflow

Step 2.5 - Add OpenAI(Message a model)



[VIEW QUESTION](#)



 PAUSE N8N LAB

 63h:28m:09s

Decision Audit

INPUT

→|

No input data

Execute previous nodes
to view input data

Parameters

Settings

Execute step

Resource
Text

Operation
Message a Model

Model
From list GPT-4.1

Messages
Type
Text

Role
System

Prompt
ROLE:
You are a senior expense compliance auditor. Validate that each category total in the structured data can be EXACTLY derived from explicit line items + taxes in the raw bill.
ROLE: You are a senior expense compliance auditor...

Tools

OUTPUT

|→

No output data

Execute step
or set mock data

Step 2.5 - Add OpenAI(Message a model)

 [VIEW QUESTION](#)  [PAUSE N8N LAB](#) [63h:27m:59s](#)

Decision Audit

INPUT

→|

No input data

Execute previous nodes
to view input data

Parameters Settings [Execute step](#)

Y derived from explicit line items + tax
var in the raw bill

ROLE: You are a senior expense compliance auditor...

Type
Text

Role
User

Prompt

INPUTS:
1. Raw bill text:
{{ \$json.raw_bill }}

2. Structured expense data to validate:
{{ \$json.structured_bill }}

Add Message

Simplify Output
☒

Connect your own custom n8n tools to this node on the
Tools

OUTPUT

|→

No output data

Execute step
or set mock data



Step 2.6 - Add Edit Fields(Set)

VIEW QUESTION

PAUSE N8N LAB

63h:27m:47s

Personal / Structure Bill

Editor Executions Evaluations

```
graph LR; A[When Executed by Another Workflow] --> B[Structure Bill  
Response Text]; B --> C[Parse Structured  
manual]; C --> D[Decision Audit  
Response Text]; D --> E[ ];
```

What happens next?

edi

Edit Image
Edits an image like blur, resize or adding border and text

Redis

Edit Fields (Set)
Modify, add, or remove item fields

Medium

LinkedIn

Convert to File
Convert JSON data to binary data

Reddit

Extract from File
Convert binary data to JSON

PredictLeads
Get company insights from PredictLeads*

Execute workflow



Step 2.6 - Add Edit Fields(Set)



VIEW QUESTION



PAUSE N8N LAB

63h:27m:31s

Parse Audit

INPUT

No input data

Execute previous nodes
to view input data

Parameters Settings **Execute step**

Manual Mapping

Fields to Set

raw_bill

T String

= {{ \$('Parse Structured').item.json.raw_bill }}

structured_bill

T String

= {{ \$('Parse Structured').item.json.structured_bill }}

decision

T String

= {{ JSON.parse(\$json.output).decision }}

rationale

T String

= {{ JSON.parse(\$json.output).rationale }}

Drag input fields here or Add Field

OUTPUT

No output data

Execute step
or set mock data



Step 2.7 - Add If Node

VIEW QUESTION

PAUSE N8N LAB

63h:27m:16s

Personal / Structure Bill

Editor

Executions

Evaluations

What happens next?

if

If

Route items to different branches (true/false)

Switch

Route items depending on defined expression or rules

DRIFT

Drift

Gotify

Agify

Get age prediction for a given name using the Agify.io API (free tier: 2,500 requests/month)

Apify

Triggers workflow on Apify Actor or task run events

Shopify

JWT

Edit Fields (Set)

Execute workflow

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Step 2.7 - Add If Node

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:27m:01s

IF Approved

INPUT

No input data

Execute previous nodes
to view input data

Parameters

Settings

Execute step

Conditions

/x {{ \$json.decision }}

is equal to

APPROVE

Add condition

Convert types where required

☐

Options

No properties

Add option

OUTPUT

No output data

Execute step

to view output data

I wish this node would...



Step 2.8 - Add Edit Fields(Set)



VIEW QUESTION



PAUSE N8N LAB

63h:26m:47s



Personal / Structure Bill

Publish

Save



Editor

Executions

Evaluations

What happens next?

ed

Edit Image

Edits an image like blur, resize or adding border and text

Redis

Edit Fields (Set)

Modify, add, or remove item fields

Reddit

Medium

Execution Data

Add execution data for search

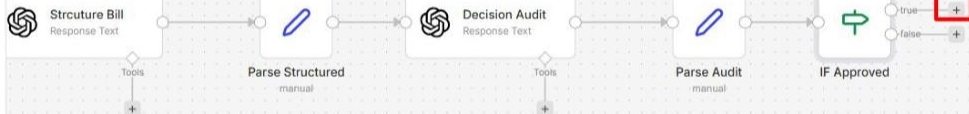
Remove Duplicates

Delete items with matching field values

Convert to File

Convert JSON data to binary data

Hedy



Execute workflow



Step 2.8 - Add Edit Fields(Set)

VIEW QUESTIONPAUSE N8N LAB63h:26m:37s

Final (Approved)

INPUT

No input data
Execute previous nodes
to view input data

ParametersSettingsExecute step

Mode

Manual Mapping

Fields to Set

result

Object

= {{ JSON.parse(\$json.structured_bill) }}

Drag input fields here or Add Field

Include Other Input Fields

Options

No properties

Add option

I wish this node would...

OUTPUT

No output data
Execute step
or set mock data



Step 2.9 - Add OpenAI(Message a model)



VIEW QUESTION



PAUSE N8N LAB

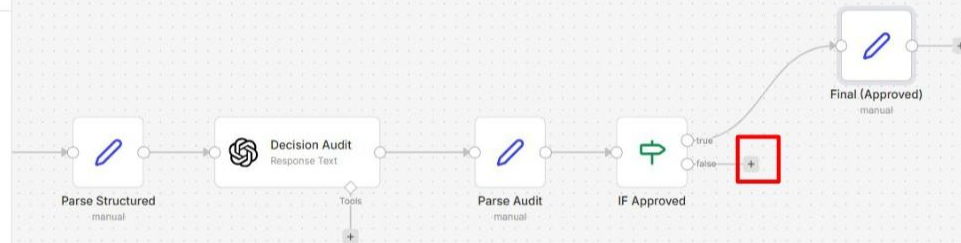
63h:26m:05s

Publish

Save



Editor Executions Evaluations



OpenAI

Search OpenAI Actions...

Actions (16)

TEXT ACTIONS

Message a model

Classify text for violations

IMAGE ACTIONS

Analyze image

Generate an image

Edit image

AUDIO ACTIONS

Generate audio

Transcribe a recording

Translate a recording

FILE ACTIONS

Delete a file

Execute workflow



Step 2.9 - Add OpenAI(Message a model)



[VIEW QUESTION](#)



[PAUSE N8N LAB](#)

[63h:25m:46s](#)

Restructure Bill

INPUT

→|

No input data

Execute previous nodes
to view input data

Parameters

Settings

Execute step

Text

Operation

Message a Model

Model

From list

GPT-4.1

Messages

Type

Text

Role

System

Prompt

ROLE:
You are a senior Corporate Expense Anal
yst. You are correcting a previously BL
OCKED expense structure. Fix ONLY the i
ssues called out by the auditor; do no
t alter correct fields.
ROLE: You are a senior Corporate Expense Analyst...

Tools

+

OUTPUT

|→

No output data

Execute step
or set mock data

Step 2.9 - Add OpenAI(Message a model)

 **Great Learning**

VIEW QUESTION

PAUSE N8N LAB

63h:25m:37s

Restructure Bill

INPUT

No input data

Execute previous nodes
to view input data

Parameters

Settings

Execute step

ROLE: You are a senior Corporate Expense Analyst...

Type

Text

Role

User

Prompt

INPUTS:

1. Raw bill text:
{{ \$json.raw_bill }}

2. Previously structured (BLOCKED):
{{ \$json.structured_bill }}

Add Message

Simplify Output

Connect your own custom n8n tools to this node on the canvas

Tools

OUTPUT

No output data

Execute step
or set mock data



Step 2.10 - Add Edit Fields(Set)

VIEW QUESTION

PAUSE N8N LAB

63h:25m:25s

Personal / Structure Bill

Publish Save

Editor Executions Evaluations

```
graph LR; A[Parse Structured manual] --> B[Decision Audit Response Text]; B --> C[Parse Audit manual]; C --> D[IF Approved]; D -- true --> E[Final (Approved) manual]; D -- false --> F[Restructure Bill Response Text]; F --> G[ ]
```

What happens next?

- Edit Image
Edits an image like blur, resize or adding border and text
- Redis
- Edit Fields (Set)**
Modify, add, or remove item fields
- Reddit
- Medium
- Execution Data
Add execution data for search
- Remove Duplicates
Delete items with matching field values
- Convert to File
Convert JSON data to binary data
- Hedy

Execute workflow

Step 2.10 - Add Edit Fields(Set)

Great Learning

 VIEW QUESTION



 PAUSE N8N LAB

 63h:25m:07s

Final (Restructured)

INPUT

No input data

Execute previous nodes
to view input data

ParametersSettingsExecute step

Mode

Manual Mapping

Fields to Set

result

Array

=

{{ typeof \$json.output === 'string' ? JSON.pa
rse(\$json.output) : \$json.output }}

Drag input fields here or Add Field

Include Other Input Fields

Options

No properties

Add option

I wish this node would...

OUTPUT

No output data

Execute step
or set mock data



Step 2.11 - Save and Publish the Sub Workflow

VIEW QUESTION

PAUSE N8N LAB

63h:24m:50s

Personal / Structure Bill

Editor Executions Evaluations

Publish

Save

Once workflow prepared with all the required Nodes, Click on Save and Publish the Workflow

Execute workflow

Workflow - 3 : MCP Server Workflow

 [VIEW QUESTION](#)



⏸ PAUSE N8N LAB

63h:23m:18s

+

□

Personal

mcp_server_trigger

Saved



Editor

Executions

Evaluations



Add first step...





Step 3.2 - Add MCP Server Trigger

VIEW QUESTION

PAUSE N8N LAB

63h:23m:06s

Personal / mcp_server_trigger

Publish

Saved

Editor

Executions

Evaluations

What triggers this workflow?

A trigger is a step that starts your workflow

MCP Server Trigger
Expose n8n tools as an MCP Server endpoint

MCP Client

Mailchimp

Microsoft SharePoint

MercadoPago
Consume MercadoPago API

MrScraper
Interact with the MrScraper API

Merge Agent Handler MCP
Connect to a Merge Agent Handler Tool Pack and call MCP tools

Microsoft Graph Security



Step 3.2 - Add MCP Server Trigger

VIEW QUESTION

PAUSE N8N LAB63h:22m:46s

MCP Server Trigger

Listen for MCP events

Execute step

Once you've finished building your workflow, run it without having to click this button by using the production URL. [More info](#)

When will this node trigger my flow? ▾

ParametersSettingsExecute step

MCP URL

Test URLProduction URL

http://localhost:5678/mcp-test/test-server/sse

Authentication

None

Path

test-server

I wish this node would...

Tools

OUTPUT

No trigger output

Test this trigger

or set mock data

Step 3.3 - Add Call n8n Workflow Tool

 **Great Learning**

VIEW QUESTION

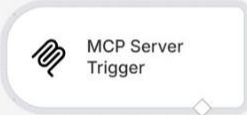
PAUSE N8N LAB

63h:22m:15s

Personal / mcp_server_trigger

Publish Save

Editor Executions Evaluations

 
Tools


Tools

Recommended Tools

 **Call n8n Workflow Tool**
Uses another n8n workflow as a tool. Allows packaging any n8n node(s) as a tool.

Other Tools

 **Calculator**
Make it easier for AI agents to perform arithmetic

 **Google Calendar Tool**
Consume Google Calendar API

 **LoneScale Tool**
Create List, add / delete items

 **TimescaleDB Tool**
Add and update data in TimescaleDB

 **Google Cloud Natural Language Tool**
Consume Google Cloud Natural Language API

 **Google Cloud Realtime Database Tool**
Interact with Google Firebase Realtime Database





Step 3.3 - Add Call n8n Workflow Tool

Great Learning

VIEW QUESTION

PAUSE N8N LAB

63h:21m:49s

structure_bill

INPUT Mapping From AI

? Parent nodes

No input connected

Parameters Settings Execute step

See an example of a workflow to suggest meeting slots using AI [here](#).

Description

Convert raw bill text into a clean structured JSON with category-wise (food/travel/hotel/others) tax-inclusive totals. Internally runs a structuring agent, an arithmetic

This tool will call the workflow you define below, and look in the last node for the response. The workflow needs to start with an Execute Workflow trigger

Source

Database

Workflow

From list Sub: Structure Bill

Workflow Inputs

raw_bill

fx {{ \$fromAI('raw_bill', '', 'string') }}

Add workflow input to send

I wish this node would...

OUTPUT

Click on the **Dropdown** list and select the published sub workflow

Note: If sub workflow not published, we can't see the sub workflow option in this list

No output data

Output will appear here once the parent node is run



Step 3.4 - Add Execute Command

VIEW QUESTION

PAUSE N8N LAB

63h:21m:35s

Personal / mcp_server_trigger

Editor

Executions

Evaluations

What happens next?

comm

>_

Execute Command

Executes a command on the host

Woo WooCommerce

Customer Messenger (n8n training)

YouTube Comments Downloader

Microsoft Dynamics CRM

Klarden DATEVconnect: Order Management

Klarden DATEVconnect: Document Management

Woo WooCommerce Tool

Execute workflow



Step 3.4 - Add Execute Command



VIEW QUESTION



PAUSE N8N LAB

63h:21m:19s

> get_reimbursement_caps

Docs

INPUT Mapping From AI

Parent nodes

Parameters Settings

Execute step

OUTPUT

Description

Return the per-category reimbursement caps (food, travel, hotel) for an employee based on their company position.

Execute Once



Command

```
EMP_ID="{{ $fromAI('emp_id', 'Employee ID lik  
e E001', 'string') }}" python3 << 'PYEOF'  
import sqlite3, json, os  
  
emp_id = os.environ.get("EMP_ID", "").strip()  
if not emp_id:
```

No input connected

|→

No output data

Output will appear here once the
parent node is run

I wish this node would...



Step 3.5 - Add Execute Command



VIEW QUESTION



PAUSE N8N LAB

63h:21m:04s

Personal / mcp_server_trigger

Publish

Save



Editor

Executions

Evaluations

What happens next?

comm



Execute Command

Executes a command on the host



WooCommerce



Customer Messenger (n8n training)



YouTube Comments Downloader
Download comments from YouTube videos,
shorts, channels, etc.



Microsoft Dynamics CRM



Klarden DATEVconnect: Order
Management
Interact with order management resources



Klarden DATEVconnect: Document
Management
Interact with document management resources
/ DMS



WooCommerce Tool

Execute workflow

Step 3.5 - Add Execute Command



VIEW QUESTION



PAUSE N8N LAB

63h:20m:49s

>_ get_past_utilization

INPUT

Mapping

From AI

? Parent nodes

No input connected

Parameters

Settings

Execute step

Description

Return the sum of an employee's already-claimed reimbursements per category for the current policy period.

Execute Once

☒

Command

```
EMP_ID="{{ $fromAI('emp_id', 'Employee ID lik  
e E001', 'string') }}" python3 << 'PYEOF'  
import sqlite3, json, os  
  
emp_id = os.environ.get("EMP_ID", "").strip()  
if not emp_id:
```

OUTPUT

No output data
Output will appear here once the parent node is run

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Step 3.6 - Add Send Email



VIEW QUESTION



PAUSE N8N LAB

63h:18m:54s

Publish

Save



Editor

Executions

Evaluations

What happens next?

smtp

Send Email

Smartbooks API

Interact with Smartbooks API

DataForSEO Merchant API

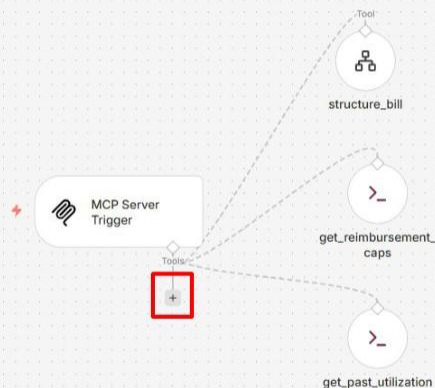
DataForSEO

DataForSEO AI Optimization API

DataForSEO

DataForSEO Domain Analytics API

DataForSEO Domain Analytics API



Execute workflow



Step 3.6 - Add Send Email



VIEW QUESTION



PAUSE N8N LAB

63h:18m:38s

send_email

INPUT

Mapping From AI

? Parent nodes

No input connected

Parameters Settings

Execute step

Credential to connect with

SMTP account 11

Tool Description

Set Automatically

Operation

Send

From Email

praneethsanthosh555@gmail.com

To Email

praneeth78211@gmail.com

Subject

`{{ $fromAI('Subject', '', 'string') }}`

Email Format

Text

Text

`{{ $fromAI('Text', '', 'string') }}`

Options

No properties

OUTPUT

No output data

Output will appear here once the parent node is run



Step 3.7 - Save and Publish the Workflow

VIEW QUESTION

PAUSE N8N LAB

63h:18m:25s

Personal / mcp_server_trigger

Editor Executions Evaluations

```
graph LR; Trigger[MCP Server Trigger] --> structure_bill[structure_bill]; Trigger --> get_reimbursement_caps[get_reimbursement_caps]; Trigger --> get_past_utilization[get_past_utilization]; Trigger --> send_email[send_email];
```

Execute workflow

Publish Save

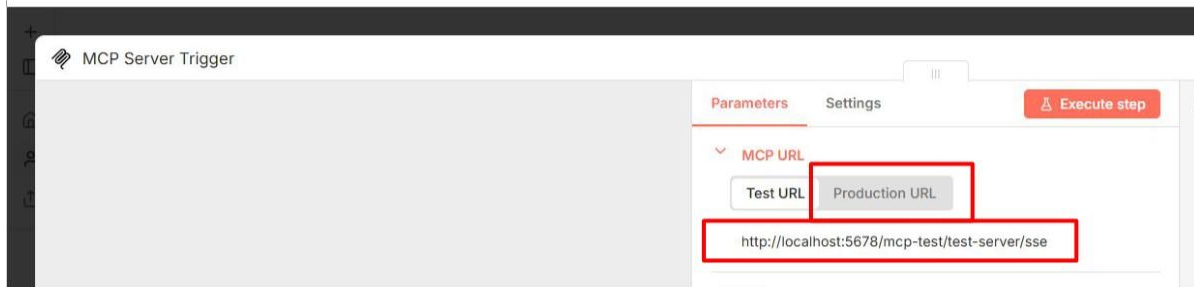
Once workflow prepared with all the required Nodes, Click on Save and Publish the Workflow

Upload receipts and Automate reimbursement

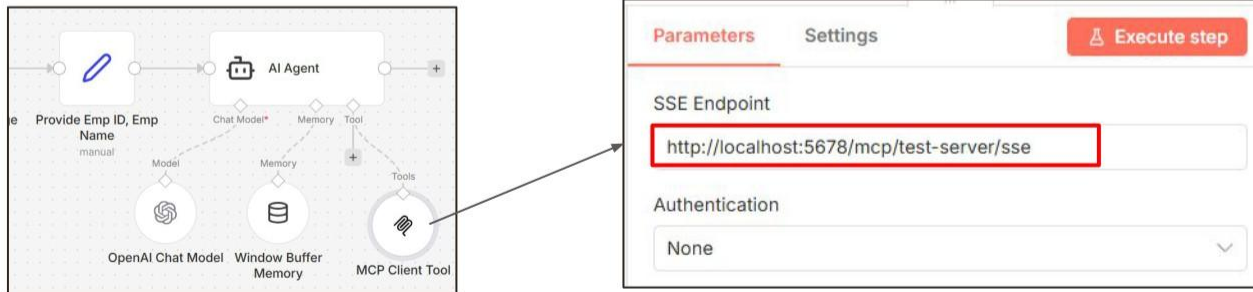


General Guidelines (Must Follow)

- Navigate to **Sub Workflow** - Open the **MCP Server Trigger** - Copy the **Production URL**, not Test URL



- Navigate to **Single Agent Workflow** - Open the **MCP Client Tool** - Paste the **Production URL** in the SSE Endpoint



- Upload the [reimbursement.db](#) database provided in to n8n lab before executing the workflows



How to Invoke the Single Agent Workflow

File - e001_Amit_Sharma.png

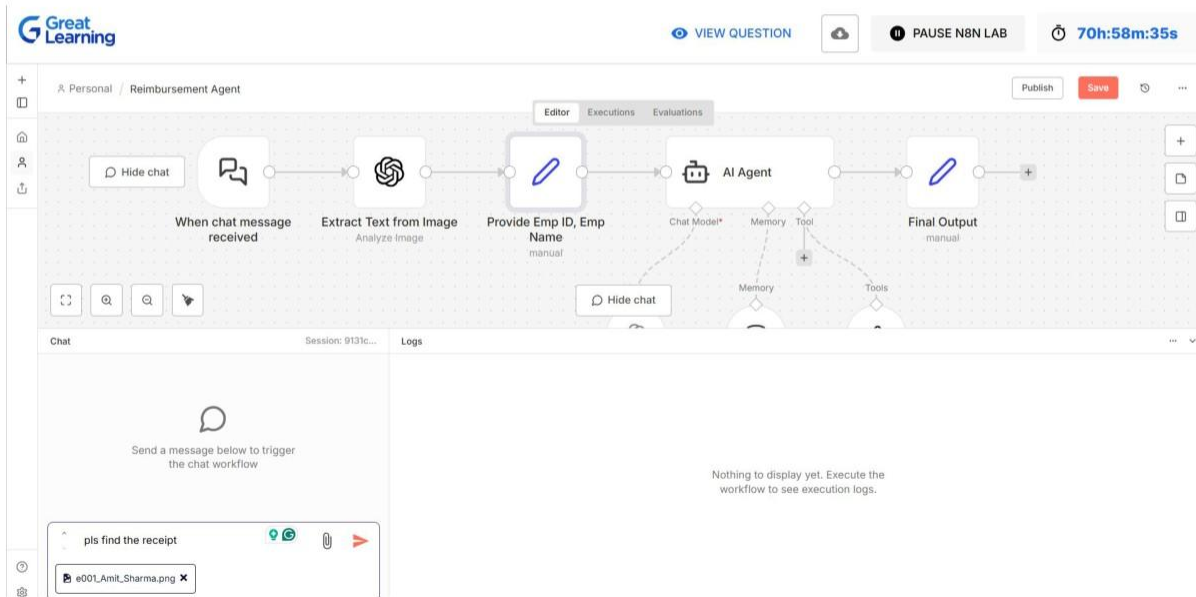
- Edit the EmpID with **E001** and emp_name with **Amit Sharma** as per the file name
- Edit the **receiver_email** to receive the reimbursement mail

The screenshot displays the Great Learning N8N LAB interface. At the top, there's a header with the Great Learning logo, a 'VIEW QUESTION' button, a 'PAUSE N8N LAB' button, and a timer showing '70h:52m:19s'. The main workspace is titled 'Provide Emp ID, Emp Name'. On the left, the 'INPUT' section shows a workflow starting with 'Extract Text from Image', followed by a loop '0' containing nodes for 'id', 'type', 'status', 'content', 'content[0]', 'type', 'annotations', 'logprobs', 'text', and 'role'. Below this is a trigger 'When chat message received' and a section for 'Variables and context'. The central 'Parameters' tab is active, showing 'Mode' set to 'Manual Mapping'. Under 'Fields to Set', two fields are highlighted with red boxes: 'Emp_ID' (set to 'E001') and 'emp_name' (set to 'Amit Sharma'). Below these, 'raw_bill' is set to a JSON path. At the bottom, another field 'receiver_email' is also highlighted with a red box. The right side of the interface shows the 'OUTPUT' section, which currently displays 'No output data' and an 'Execute step' button.



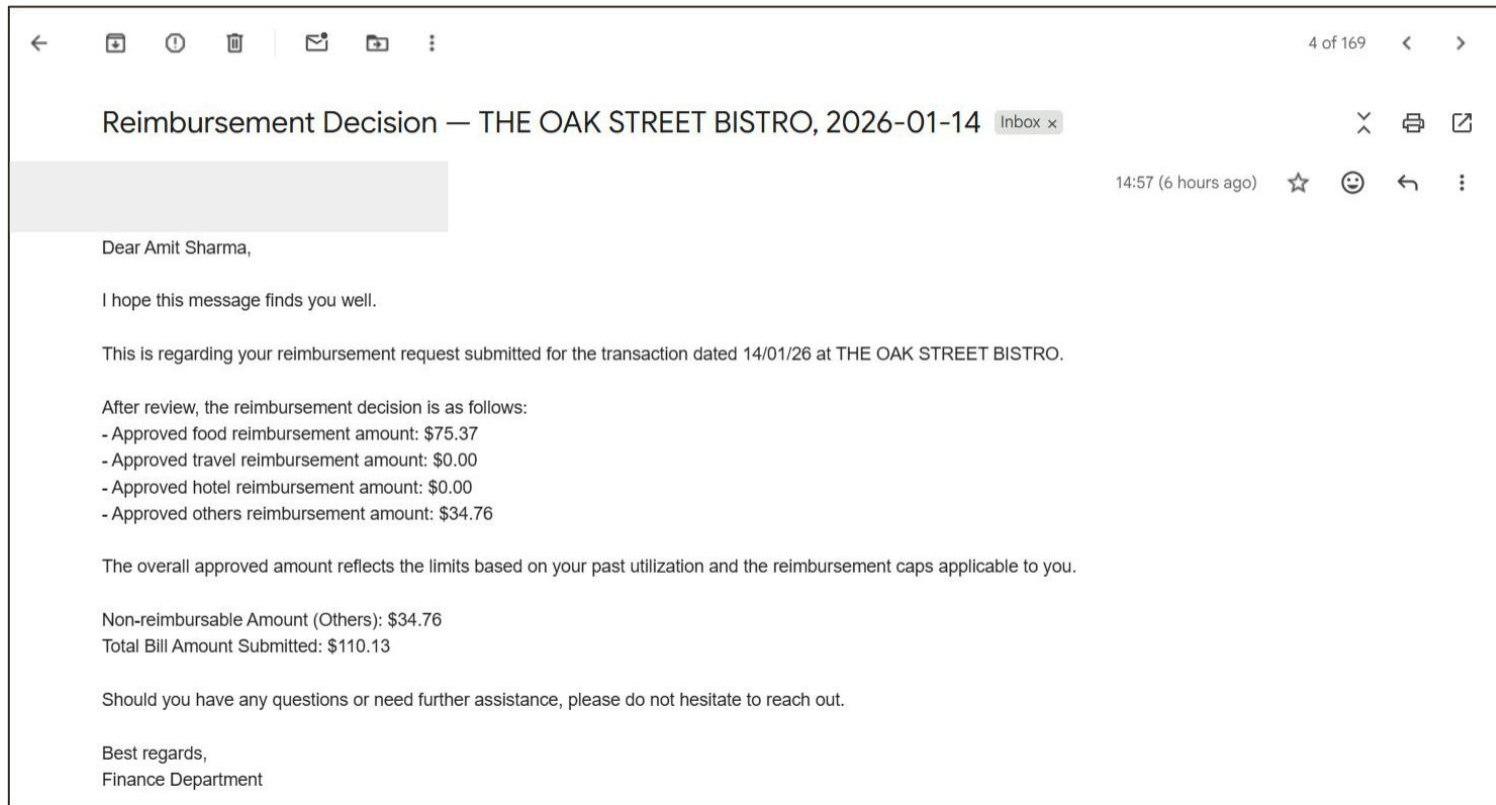
How to Invoke the Single Agent Workflow

- Upload the receipt along with a string “please find the receipt” using the Chat Option
- Click on send
- Wait for the workflow completion - You'll receive a reimbursement email once workflow completed





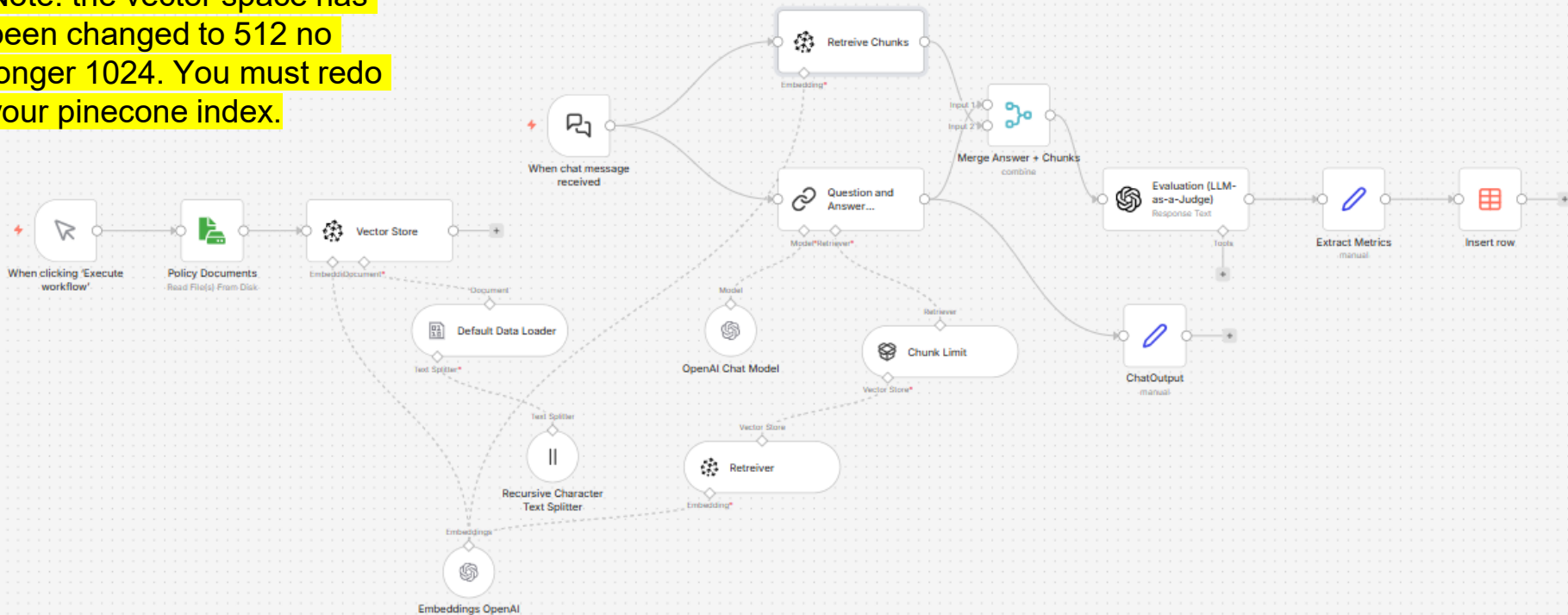
Sample Email Received





Fixed Week 5 Solution

Note: the vector space has been changed to 512 no longer 1024. You must redo your pinecone index.



https://github.com/eaglelandsonce/FixedWeek5Solution_n8n

Summary

Session Structure

- Review objectives, expectations, and session flow

Conceptual Recap

- Clarify key foundational concepts & additional explanation

Business Case Study

- Examine challenges in manual expense reimbursement
- Define an automated, policy-compliant, and auditable solution
- Build a three-workflow agentic system to:
 - Structure expense bills
 - Validate expenses against policy
 - Calculate approved and rejected amounts
 - Communicate reimbursement decisions

Guided Hands-On Demonstration

- Process bill images with GPT-4o vision to extract, categorize, and total expenses.
- Build an audit and self-correction loop to identify and resolve errors.
- Expose reimbursement workflows as MCP tools and connect the AI agent through SSE.
- Retrieve policy data, calculate reimbursement decisions, and send the final email.

Questions and Answers

Address follow-up and parked questions



Power Ahead!



APPENDIX

Case Study - Insurance Policy Q&A for Customer Support Agents

Business Context



MeridianHealth Insurance is a mid-sized health insurance carrier headquartered in Hartford, Connecticut, licensed to operate across 22 states with concentrated presence in the Northeast, Mid-Atlantic, and Midwest. The company serves approximately 2.4 million covered lives across a portfolio that spans:

- **Individual and family ACA marketplace plans** across Bronze, Silver, Gold, and Platinum metal tiers
- **Medicare Advantage and Medicare Supplement plans** for the senior segment
- **Small group and mid-market employer plans** sold through brokers and benefits consultants
- **Voluntary supplemental products** including critical illness, hospital indemnity, and accident plans

MeridianHealth booked roughly \$4.1 billion in earned premium last fiscal year and operates through a hybrid distribution model — a direct sales force of close to 4,800 licensed sales representatives, a network of 28,000 appointed independent agents and brokers, and 14 regional employer-group partnerships. The product portfolio currently spans 47 active plan designs, each with its own combination of deductibles, coinsurance, copay structures, out-of-pocket maximums, prior-authorization rules, formulary tiers, and in-network provider arrangements that vary by state and rating area.

The problem is that **sales representatives cannot reliably retrieve this information in real time** during a live conversation. New representatives, who form roughly 35% of the field force at any given time given an industry attrition rate of 28–30%, often do not know where the answer lives, let alone what it says. Even experienced representatives struggle when:

- The question crosses two or more products
- The clause involves a recent plan-year update or rider change
- The answer is buried deep in a 100-page Evidence of Coverage document



The most common workarounds - calling a back-office product helpdesk, putting the customer on hold, or making a confident guess - each carry a cost. The product helpdesk receives an estimated 11,000-13,000 internal queries per month from the field, with average resolution times of 6-14 minutes per query. When the representative guesses or paraphrases, the risk shifts to **misrepresentation**, which has become a heightened regulatory concern under tightened state market-conduct examinations and CMS oversight of Medicare Advantage marketing.

The operational and human consequences are significant:

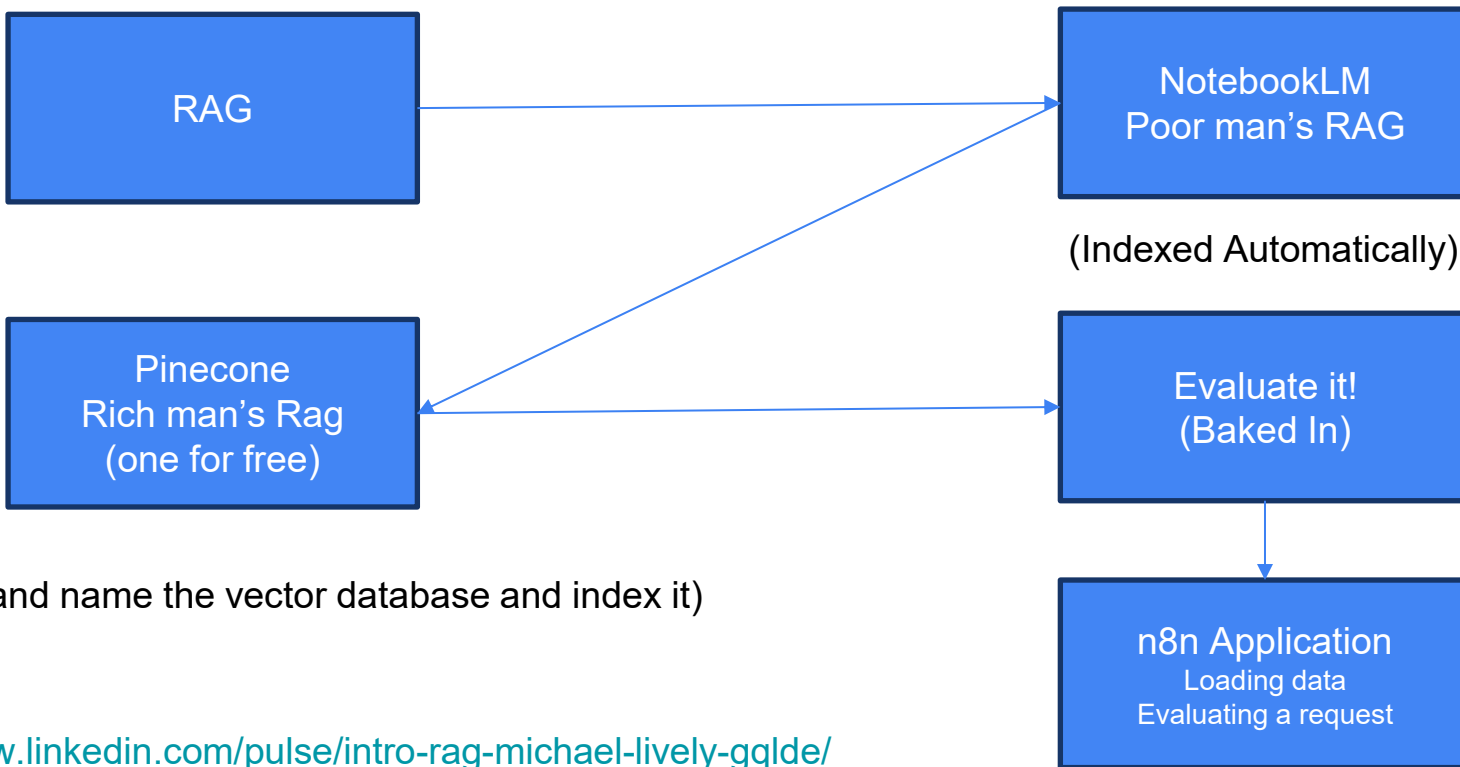
- An estimated **18–22% of sales conversations stall or drop off** at the question-handling stage, with internal win-loss reviews indicating "couldn't answer customer's clarification" as a recurring loss reason.
- Misrepresentation complaints - where the customer alleges they were told a benefit existed that the plan does not actually offer - contributed to roughly **\$1.7 million in claim disputes, goodwill payouts, and remediation costs** last year, and triggered three separate state Department of Insurance market-conduct inquiries.
- New sales representatives take **9–12 months to reach full productivity**, largely because product mastery across 47 plan designs is built through experience rather than accessible reference material.
- Senior product specialists in the back office spend an estimated **40–55% of their working hours** answering repetitive field queries, leaving little capacity for actual underwriting support and product enablement.
- The sales representative on the front line carries the cognitive and emotional load of being asked questions they cannot confidently answer in front of a customer who is deciding whether to trust them with their family's health coverage.

Meridian Health is initiating a proof-of-concept to build an in-conversation policy Q&A capability that allows a sales representative to surface accurate, source-grounded answers from across the company's product documentation in real time, and to measure the answer quality rigorously enough to deploy it in a regulated environment.

- ❖ **Reads a sales representative's question - typed or spoken - and produces a precise, grounded answer drawn from the relevant Evidence of Coverage documents, Summaries of Benefits, and approved product collateral**, so that the representative can respond to the customer within the flow of the conversation rather than putting them on hold or escalating to the back office.
- ❖ **Returns the exact source clause and document reference alongside every answer**, so that the representative can read or paraphrase from the verified wording, and so that the company has a clear audit trail of what the system told the field
- ❖ **Handles cross-product and comparative questions** - such as differences between two plan variants, rider stacking rules, or how a clause changed in the latest plan year - so that the representative can guide a customer toward the right product rather than defaulting to the one they know best.
- ❖ **Includes a built-in evaluation framework that continuously measures answer accuracy, source-faithfulness, and refusal behaviour on out-of-scope or unsafe questions**, so that product and compliance teams can quantify how trustworthy the system is before each rollout phase, and can catch regressions when plan documents or product rules are updated.



Review



(Create and name the vector database and index it)

<https://www.linkedin.com/pulse/intro-rag-michael-lively-gqlde/>

Great Learning
POWER AHEAD



⌚ 39h:32m:38s

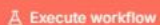
Publish

Saved

— — —

Executions

Evaluations



Open chat

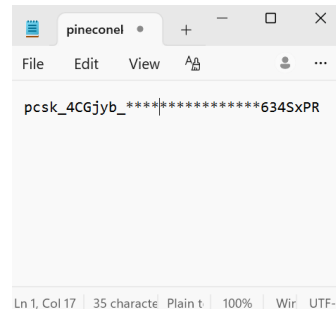
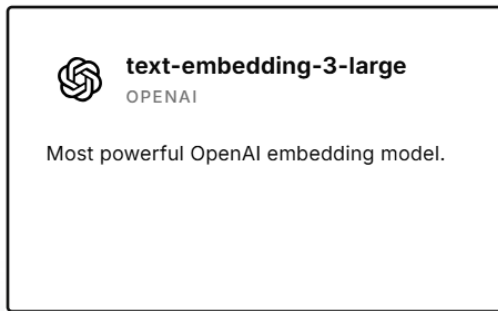
Step 1: Signup and Login

<https://www.pinecone.io/>

Step 2: Create an index (1024) Call it insurepolicy

Step 3: Create an API Key Save to a text file.

 **API keys**



n8n

6/26/2026

Michael Lively

pcsk_4CGjyb_*****

All

...

<https://www.pinecone.io/>

n8n Login

Lab Access Guidelines

To ensure that the N8N labs launch smoothly, please follow the guidelines below before starting your lab session.

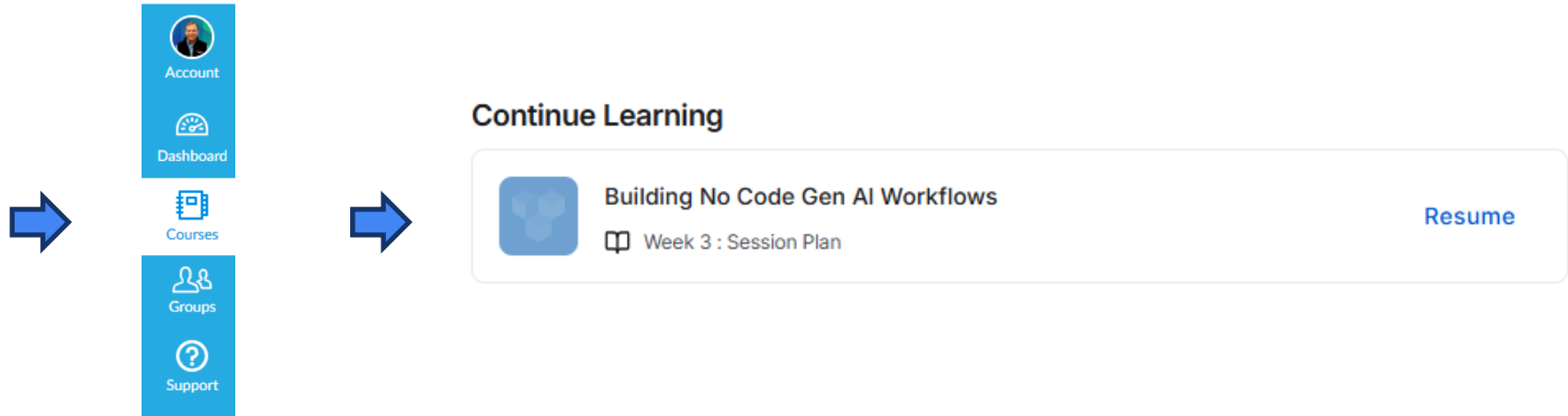
1. Access the Lab from Olympus

Always launch the lab directly from the **Olympus lab page**.

- Log in to Olympus using your Great Learning credentials.
- Navigate to the course module.
- Open the lab from the **Open link** button as shown in the below screenshot.

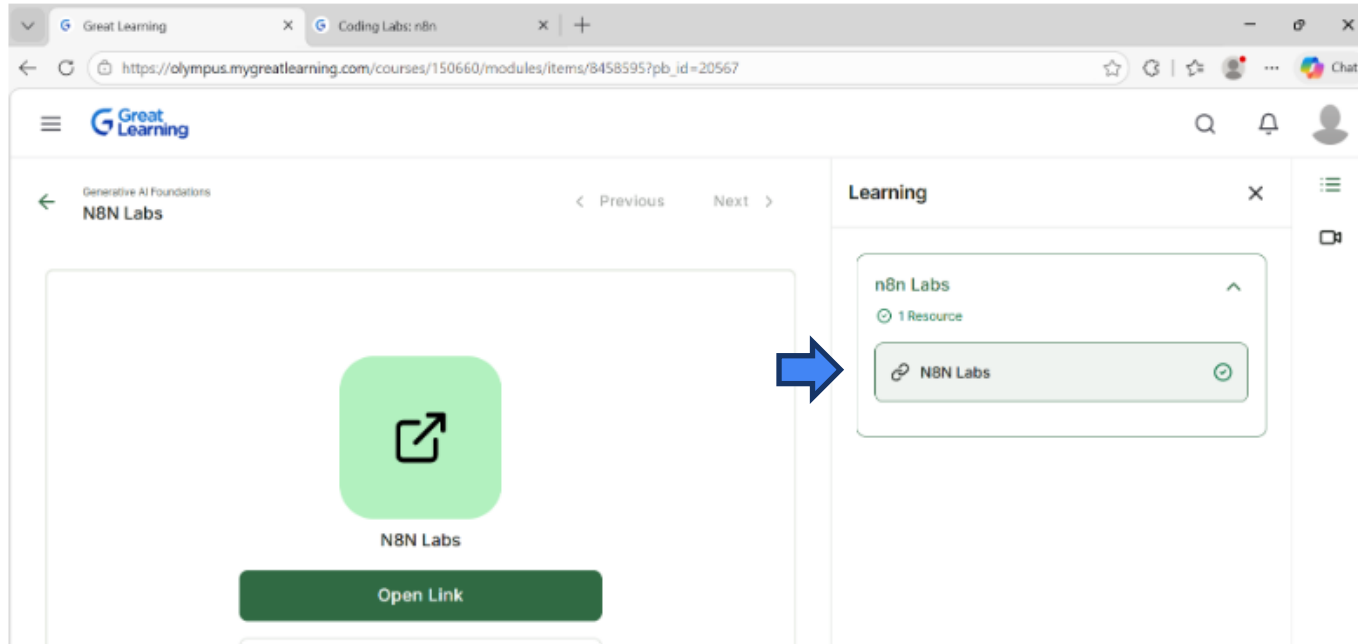
https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

n8n Login



https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

n8n Login



https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

n8n Login

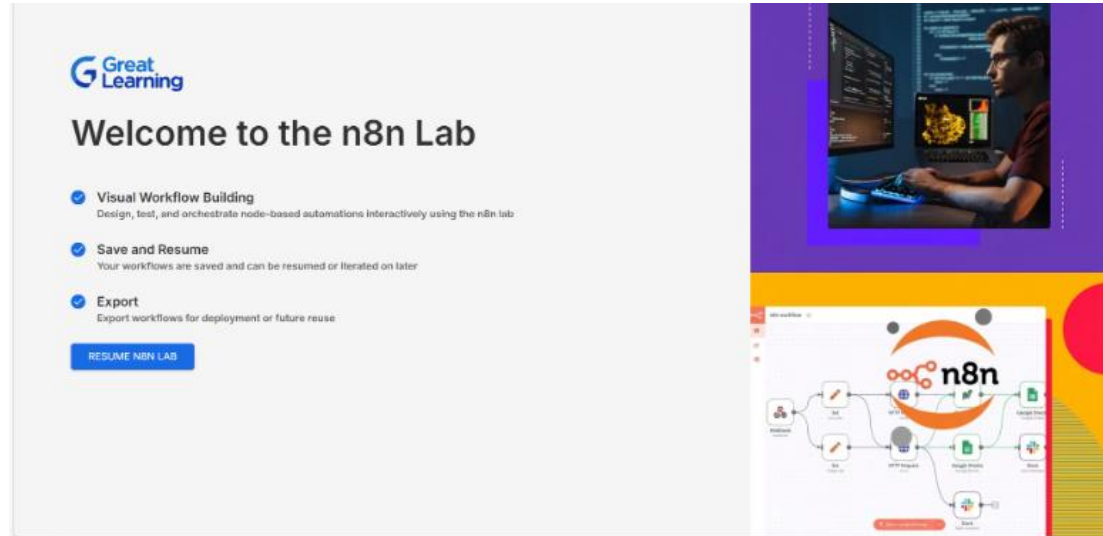
2. Use the Same Browser

When launching the lab:

- Stay logged into Olympus.
- Open the lab **in the same browser window**
- when you click on the **Open Link**(check above screenshot), the lab will open in a new tab in the same window.

https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

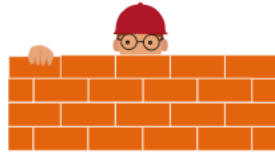
n8n Login



The image shows a composite of two parts. The left part is a 'Welcome to the n8n Lab' page from Great Learning. It features the Great Learning logo, the title 'Welcome to the n8n Lab', and three bullet points: 'Visual Workflow Building' (Design, test, and orchestrate node-based automations interactively using the n8n lab), 'Save and Resume' (Your workflows are saved and can be resumed or iterated on later), and 'Export' (Export workflows for deployment or future reuse). A blue button labeled 'RESUME N8N LAB' is at the bottom. The right part is a vertical strip containing two images: the top one shows a person working at a computer with multiple monitors, and the bottom one shows a screenshot of an n8n workflow diagram with various nodes like 'Webhook', 'JSON', 'HTTP Request', 'HTTP Response', 'Merge', 'Split', 'Delay', and 'Code'.

https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

n8n Login



0%

Creating n8n Lab, this could take upto 4 minutes...



No Incognito Mode

Do not use n8n Lab in incognito mode else certain operations will not work.

https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

n8n Login

Zoom out!



Unsupported device resolution, we recommend a minimum of 1280X768 for best experience

https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

n8n Login



 VIEW QUESTION



PAUSE N8N LAB

 31h:30m:05s



Overview

All the workflows, credentials and data tables you have access to

Create workflow

Workflows

Credentials

Executions

Variables

Data tables

Search

Sort by last updated



Evaluating GenAI Workflows

Last updated 6 hours ago | Created 19 June

Personal



Nima ABSA Workflow V2

Last updated 13 hours ago | Created 19 June

Personal



Data Analysis Agent

Last updated 3 months ago | Created 7 March

Personal



Scheduled Joke Teller

Last updated 3 months ago | Created 7 March

Personal



Chat Joke Teller

Last updated 3 months ago | Created 7 March

Personal



Research Agent

Last updated 3 months ago | Created 14 February

Personal



Story Writer Looping Demo

Last updated 3 months ago | Created 5 March

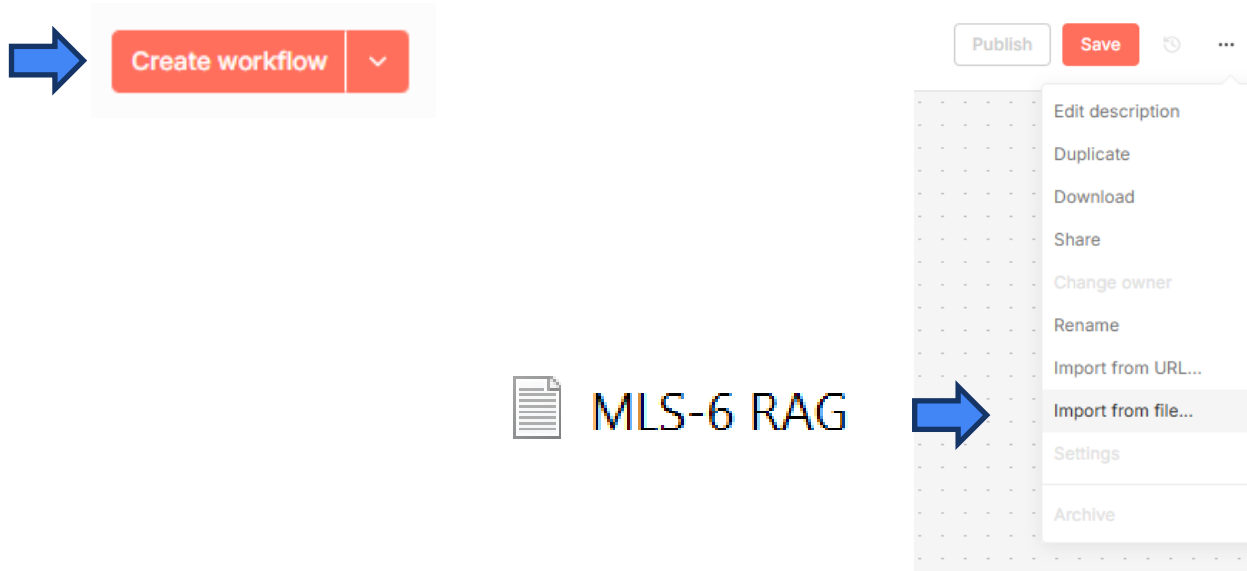
Personal



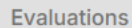
Quote Teller

https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550

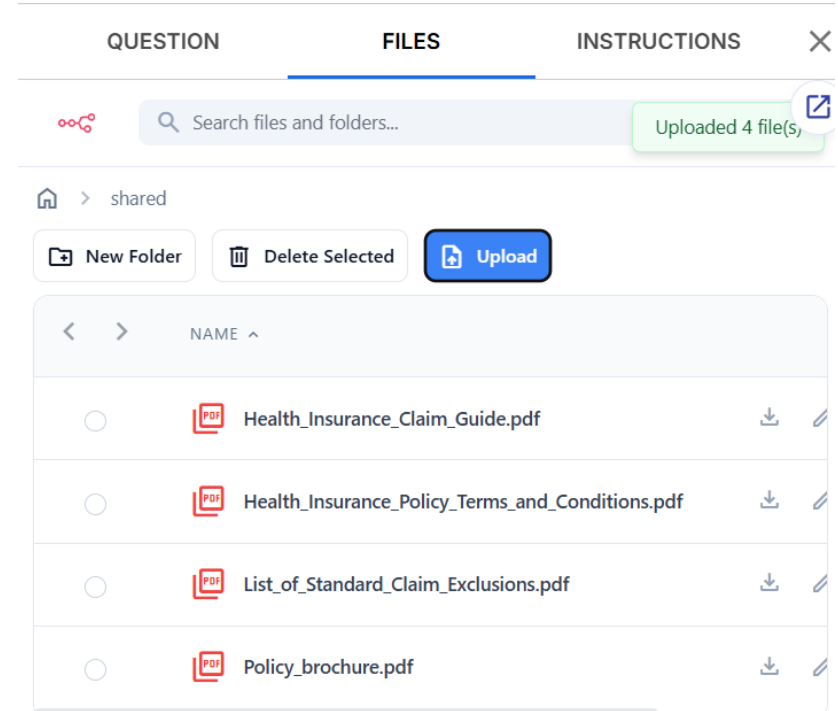
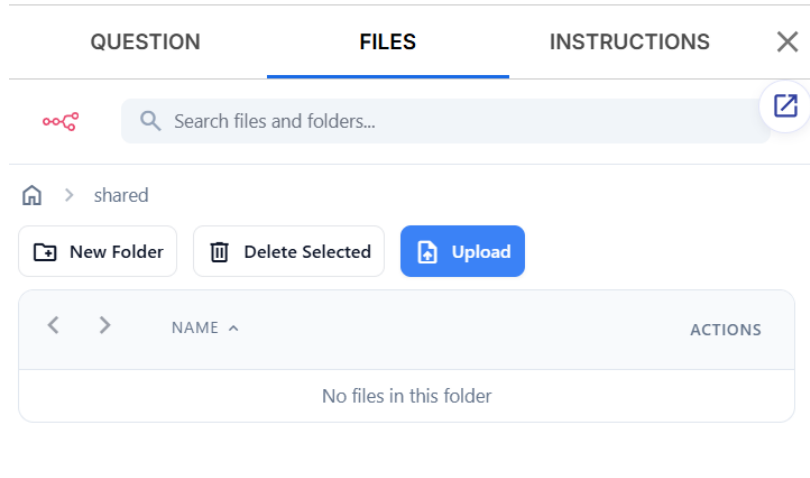
n8n Login



https://olympus.mygreatlearning.com/courses/152925/files/13775388?module_item_id=8826550



Import Files



Add Your API Key



Credential to connect with

PineconeApi account 1394

 **PineconeApi account 1394** 

 PineconeApi

Connection

Sharing

Details

 Connection tested successfully 

Need help filling out these fields? [Open docs](#)

API Key *
.....

Allowed HTTP Request Domains
All 

Test Cases - Queries

01. Process Timelines

If I use the cashless claim process, how long does it take for pre-authorisation and final settlement approval?

02. Domiciliary Care

Is 'Domiciliary Hospitalisation' for Asthma admissible, and are oxygen masks or nebulisers covered?

03. Newborn Inclusion

What is the earliest age to add a baby, and will the policy pay for infant formula or baby wipes?

04. Recovery Equipment

For accidental injury, will the insurer cover the cost of a CPAP machine or an arm sling?

05. Pregnancy & Childbirth

Are standard maternity expenses covered under basic plans, and are baby food, diapers, or delivery kits reimbursable?

Prerequisites – Create Data Table

Prerequisites - Navigate to Data Tables

VIEW QUESTION

PAUSE N8N LAB

39h:42m:26s

Personal

Workflows, credentials and data tables owned by you

Create workflow

Workflows

Credentials

Executions

Data tables

Search

Sort by last updated

Personal

Workflows, credentials and data tables owned by you

Create data table

Workflows

Credentials

Executions

Data tables

Create new data table

Data table name *

evaluation_results

☒ From scratch

☐ Import CSV

Cancel

Create

Prerequisites - Add Columns



[VIEW QUESTION](#)



 PAUSE N8N LAB

 39h:40m:14s

Personal / Data tables /

id

createdAt

updatedAt

+

chatinput

response

relevance

groundedness

context_precision

context_recall

verdict

Name *

chatinput

Type *

string

Add Column



Personal / Data tables /

Search

Add Row

Add Column

☐ id	T chatinput	T Answer	T relevance	T groundedness	T context_precision	T context_recall	T verdict	createdAt
-----------------------------	-------------	----------	-------------	----------------	---------------------	------------------	-----------	-----------

Step 10.9 - Add Data Table



Insert row

INPUT



Schema Table JSON

Parameters Settings

Execute step

OUTPUT



Schema Table JSON

Docs X

Extract Metrics 11 items

chatInput How to apply for cashless claim?

response **Answer** \n - Visit a hospital from the insurer's network list. \n - Notify the insurer at least 48 hours before planned admission or within 24 hours for emergency hospitalisation. \n - Fill out the pre-authorisation form at the hospital's insurance or TPA desk and send it to the insurer. \n - Present the patient's medical ID card, digital health card from the insurer's app, and the pre-authorisation letter at the hospital desk. \n - The claims management team reviews and issues an approval letter if all is in order. \n - Payment is made directly to the network provider for cashless claims, consti...

relevance 5

groundedness 5

context_precision 5

context_recall 5

verdict PASS

Message a model 11 items

Merge 11 items

Question and Answer Chain 1 item

Pinecone Vector Store1 10 items

When chat message received 1 item

Variables and context

Resource

Row

Operation

Insert

Data table

From list evaluation_result_data

Mapping Column Mode

Map Each Column Manually

Values to insert

userquery

fx {{ \$json.chatInput }}

How to apply for cashless claim?

Answer

fx {{ \$json.response }}

Answer - Visit a hospital from the insurer...

relevance

fx {{ \$json.relevance }}

5

groundedness

fx {{ \$json.groundedness }}

5

11 items

userquery [null]

Answer [null]

relevance [null]

groundedness [null]

context_precision [null]

context_recall [null]

overall_score [null]

verdict [null]

id 108

createdAt 2026-05-08T06:50:18.764Z

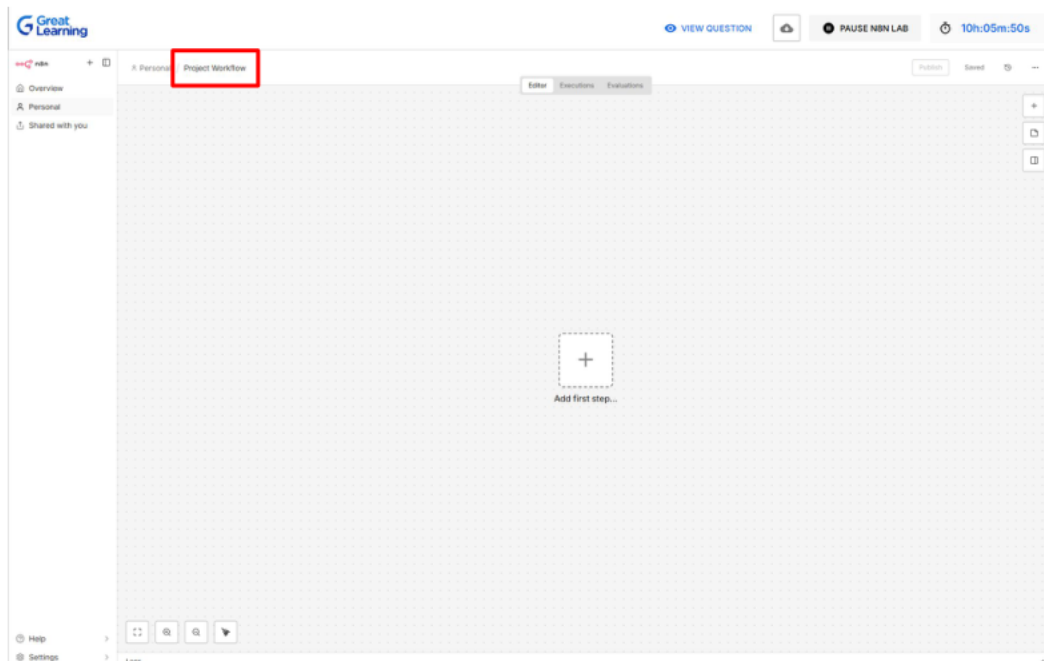
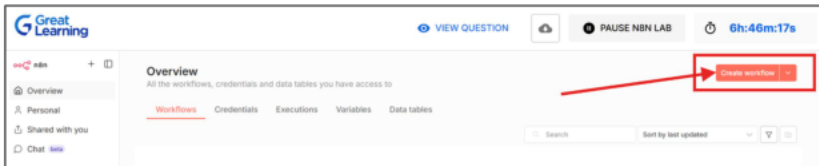
updatedAt 2026-05-08T06:50:18.764Z

Workflow 1 – Vector Indexing Workflow

Part 1 – Create a blank workflow

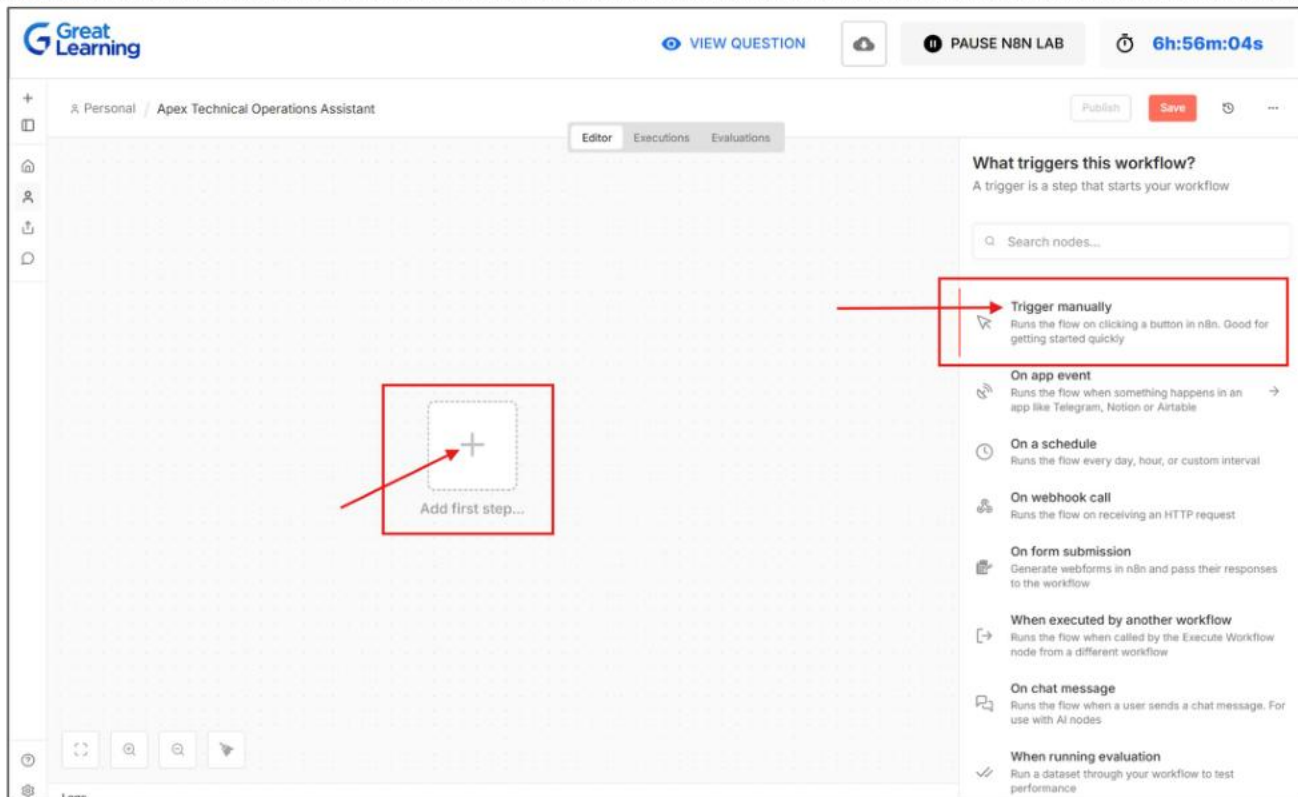
Step 1.1 - Create a New Workflow

❖ Open your n8n dashboard → Click **Create Workflow** → Rename the workflow to **“Project Workflow”**



Step 1.2 - Add Manual Trigger

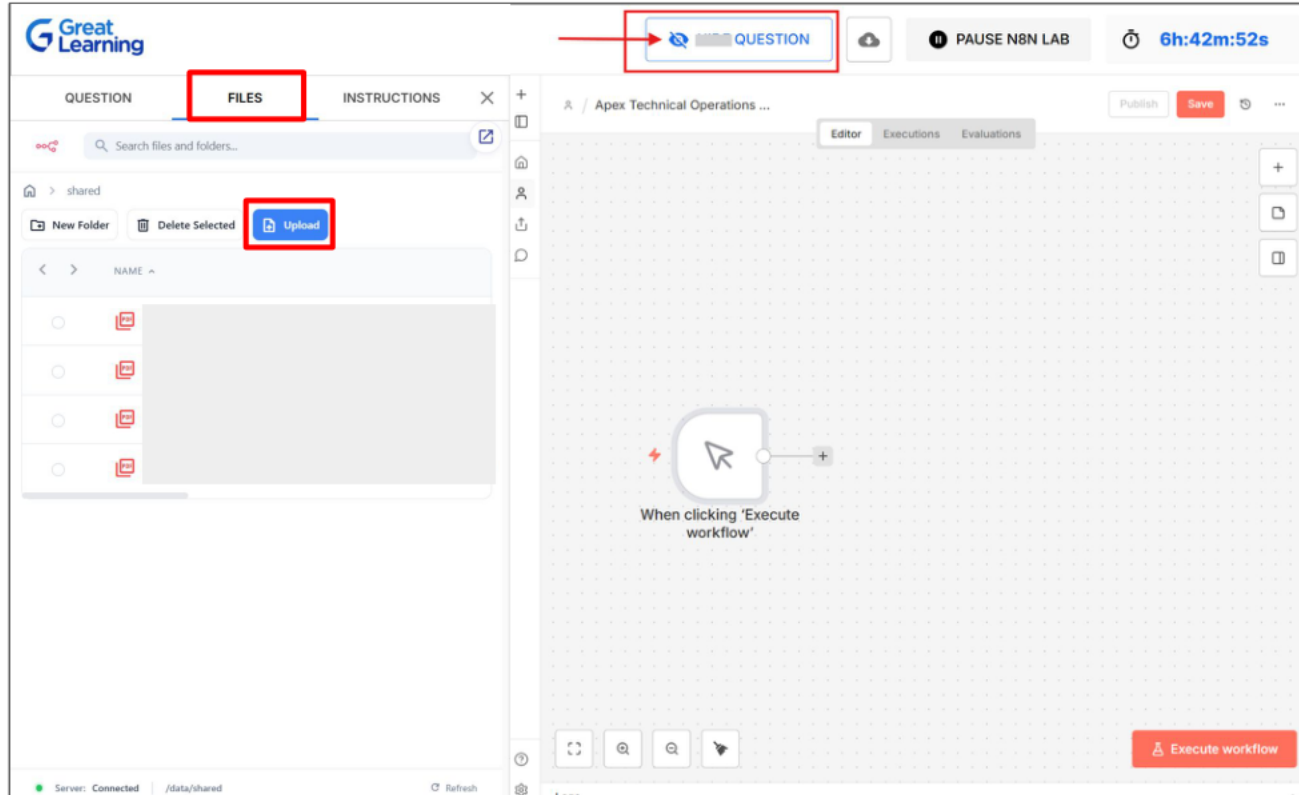
- ❖ Click “+ Add first step...” → Select “Trigger Manually” → Automatically it will add to Canvas
- ❖ This trigger allows manual execution while building and testing the workflow



Part 2 – Read and Prepare Documents

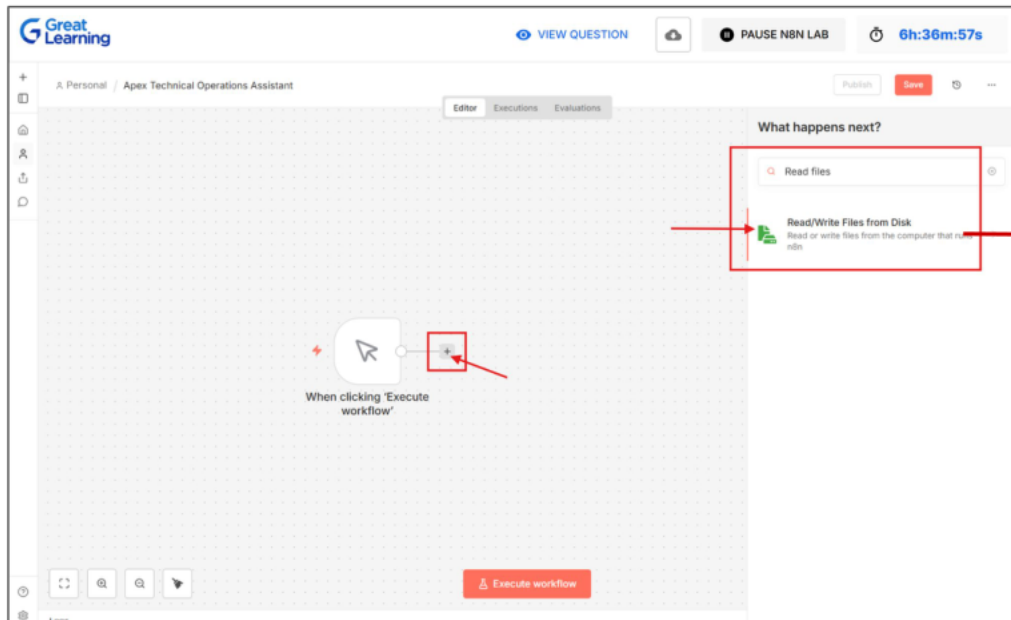
Step 2.1 - Upload Documents to n8n Workspace

- ❖ At the top of the interface, click **View Question** → A sidebar opens → Go to Files, Click on Manage Files → Upload all PDF manuals from your local system (These files represent the company's technical knowledge base)



Step 2.2 - Add “Read Files from Disk” Node

- ❖ Click + **Add Node** next to trigger → Search for **Read/Write Files from Disk** Node → Select **Read Files from Disk**



Step 2.3 - Execute “Read Files” Node

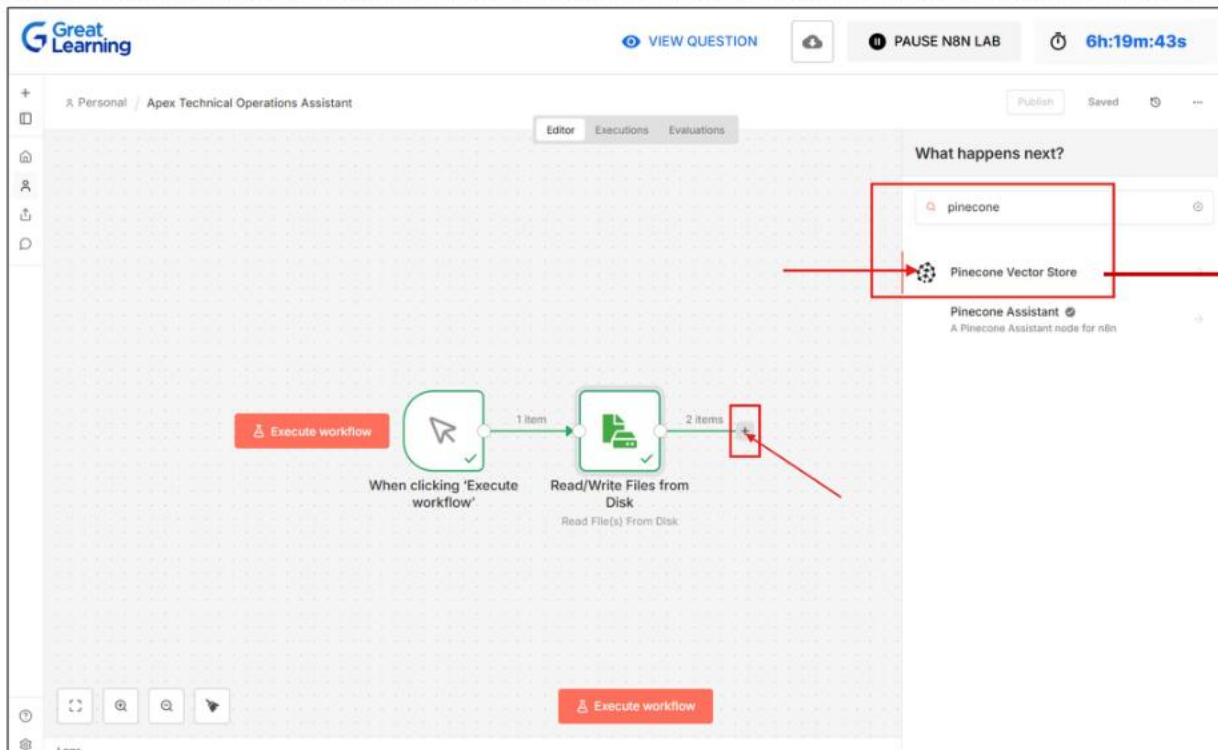
- ❖ In **File(s) Selector**, enter: `/data/shared/*`, this reads all uploaded files from the n8n workspace.
- ❖ Click on **Execute step**

The screenshot displays the n8n workflow editor. At the top, the Great Learning logo is on the left, and navigation buttons for 'VIEW QUESTION', 'PAUSE N8N LAB', and a timer '40h:53m:41s' are on the right. The main workspace is titled 'Policy Documents'. On the left, the 'INPUT' panel shows a workflow step with the instruction 'When clicking "Execute workflow"'. The central 'Parameters' panel is for the 'Read File(s) From Disk' node. It includes a description, an 'Operation' dropdown set to 'Read File(s) From Disk', a 'File(s) Selector' text field containing '/data/shared/*.pdf', and an 'Options' section with 'No properties'. A red box highlights the 'Execute step' button at the top of the parameters panel and the 'File(s) Selector' field. On the right, the 'OUTPUT' panel shows two data items. Item 1 is a PDF file named 'Health_Insurance_Claim_Guide.pdf' with a size of 290 kB. Item 2 is a PDF file named 'Health_Insurance_Policy_TandC.pdf' with a size of 568 kB. Both output items are enclosed in a red box. A 'Download' button is visible below the first output item.

Part 3 – Add Pinecone Vector Store

Step 3.1 - Add Pinecone Vector Store Node

- ❖ Click **+ Add Node** → Search for **Pinecone Vector Store** → Select operation: **Add Documents to Vector Store**
- ❖ This node stores document embeddings inside Pinecone



Step 3.2 - Pinecone Configuration Fields

- ❖ In the **Configuration** tab, you will see two required fields: **Credential to connect with** (Pinecone API key), **Pinecone Index**. In the next steps, we will create and configure both of these.

The screenshot shows the 'Pinecone Vector Store' configuration window within the Great Learning N8N Lab. The interface is divided into several sections:

- Header:** Includes the Great Learning logo, a 'VIEW QUESTION' button, a 'PAUSE N8N LAB' button, and a timer showing '6h:09m:09s'.
- Left Panel:** Labeled 'INPUT', it contains a 'Read/Write Files from Disk' button and a large empty box labeled 'Documents'.
- Configuration Panel:** Titled 'Parameters', it contains the following fields:
 - Credential to connect with:** A dropdown menu with 'Select Credential' as the current selection. This field is highlighted with a red box and a red arrow.
 - Operation Mode:** A dropdown menu.
 - Insert Documents:** A dropdown menu.
 - Pinecone Index:** A dropdown menu with 'From list' and 'Choose...' as options. This field is also highlighted with a red box and a red arrow.
 - Embedding Batch Size:** A text input field with the value '200'.
 - Options:** A section with 'No properties' and an 'Add Option' button.
- Right Panel:** Labeled 'OUTPUT', it shows 'No output data' and an 'Execute step' button, with a note 'or set mock data'.
- Footer:** Includes a tip: 'Tip: Get a feel for vector stores in n8n with our RAG starter template' and a feedback prompt: 'I wish this node would...'.

Part 4 – Create Pinecone Account and API Credential

Step 4.1 - Create Pinecone Account

- ❖ Open pinecone.io → Sign up or log in → Navigate to **API Keys**
- ❖ Copy the API Key value

The screenshot displays the Pinecone web interface. In the left-hand navigation menu, the 'API keys' option is highlighted with a red rectangular box. The main content area is titled 'API keys' and features a table with the following data:

Name	Created on	Created by	Value	Permissions	Actions
default	22/02/2026		psk_PfV5d_*****	All	...

This row in the table is also highlighted with a red rectangular box. A '+ API key' button is located in the top right corner of the table area. In the bottom left corner, there is a 'STARTER USAGE' section with the following metrics:

- WUs: 11K / 2M
- Storage: 0.0035GB / 2GB
- RUs: 2 / 1M

An 'Upgrade now' button is positioned below these metrics.

Step 4.2 - Add Pinecone Credential in n8n

- ❖ Return to Pinecone node configuration → Click **Create New Credential** → **Paste API key** → **Save and test connection**
- ❖ If successful, credential is validated automatically

Parameters Settings **Execute step**

Tip: Get a feel for vector stores in n8n with our [RAG starter template](#)

Credential to connect with

Select Credential

+ Create new credential

Pinecone Index

From list Choose...

Embedding Batch Size

200

Options

No properties

Add Option

Great Learning

VIEW QUESTION PAUSE N8N LAB 5h:19m:45s

Pinecone Vector Store

PineconeAPI account

Connection

Sharing

Details

Connection tested successfully

Need help filling out these fields? [Open docs](#)

API Key *

Allowed HTTP Request Domains

All

Save

Part 5 – Create Pinecone Index

Step 5.1 - Create New Index in Pinecone

- ❖ Go to Pinecone dashboard → Navigate to **Database**, go inside **'indexes'** → Click **Create Index**

The screenshot shows the Pinecone dashboard interface. The breadcrumb navigation at the top reads 'Pinecone / h's Org / Default / Database'. The left sidebar contains a menu with 'Indexes' highlighted by a red box. The main content area is titled 'Indexes' and features a large blue button labeled 'Create index' (also highlighted by a red box) and a smaller 'Quickstart' button. Below the button, text states 'Indexes store and search vector embeddings for AI applications.' In the bottom-left corner, there is a 'STARTER USAGE' section with a progress bar for 'WUs' (11K / 2M), 'Storage' (0.0035GB / 2GB), and 'RUs' (2 / 1M), along with an 'Upgrade now' button.

Step 5.2 - Configuring the New Index in Pinecone

- ❖ Go to Pinecone dashboard → Navigate to **Database**, go inside 'indexes' → Click **Create Index**.
- ❖ Configure: **Example Index Name:** **insurancepolicy**, **Embedding Model:** **OpenAI text-embedding-3-small**, **Dimension:** **512** (Leave remaining settings as default)

The screenshot shows the Pinecone dashboard interface. The top navigation bar includes 'Pinecone', 'Great Learning', 'Chatbot application', 'Database', 'Marketplace', 'New', 'Docs', 'Settings', 'Get help', and a user profile 'PK'. The left sidebar contains links for 'Get started', 'Database', 'Quickstart', 'Indexes (3)', 'Backups', 'Assistant', 'Inference', 'API keys', and 'Manage'. The main content area is titled 'Configuration' and shows three options: 'text-embedding-3-large' (Most powerful OpenAI embedding model), 'text-embedding-3-small' (Most cost effective OpenAI embedding model, highlighted with a red box), and 'Manual configuration' (Customize settings for your embedding model). Below the configuration options, a table shows the index configuration for 'text-embedding-3-small':

Index configured for text-embedding-3-small:	
Modality	Text
Vector type	Dense
Max input	8,191 tokens
Dimension	512
Metric	cosine

To the right of the table, a section titled 'Manage your own embeddings' provides instructions: 'This model does not support integrated inference. To manage your own embeddings, follow these steps: 1. Create an account: Sign up and obtain an OpenAI API key. 2. Embed: Generate vector embeddings for your data. 3. Upsert: Insert the generated embeddings into your index.' At the bottom left, a 'DATABASE USAGE' section shows 'RUs 7 / 1M', 'Storage 0.016GB / 2GB', and 'WUs 279K / 2M', with an 'Upgrade now' button. At the bottom right, a 'Create index' button is highlighted with a red box, next to a 'Cancel' button. A status bar at the bottom indicates 'You are currently on the Starter plan'.

Step 5.3 - Select Index in n8n

- ❖ **Return to the Pinecone node configuration.**
In the **Pinecone Index** field, you will see the index you previously created listed here.
- ❖ Select **insurancepolicy** from the **"From list"** dropdown and keep the **Embedding Batch Size** set to the default value (**200**).

The screenshot displays the n8n Vector Store configuration interface. On the left, under the 'INPUT' tab, two data sources are listed: 'Policy Documents' (4 items) and 'Health_Insurance_Policy_TandC.pdf' (568 kB). The main configuration area is divided into 'Parameters' and 'Settings' tabs. The 'Parameters' tab is active, showing a 'Pinecone Index' dropdown menu with 'From list' selected and 'insurancepolicy' chosen. This dropdown is highlighted with a red rectangle. Other visible settings include 'Credential to connect with' (PineconeApi account 461), 'Operation Mode' (Insert Documents), 'Embedding Batch Size' (200), and 'Options' (No properties). A tip box at the top suggests using a RAG starter template. The right side of the interface shows 'Output' and 'Logs' tabs, with a message indicating 'No output data' and buttons for 'Execute step' or 'set mock data'.